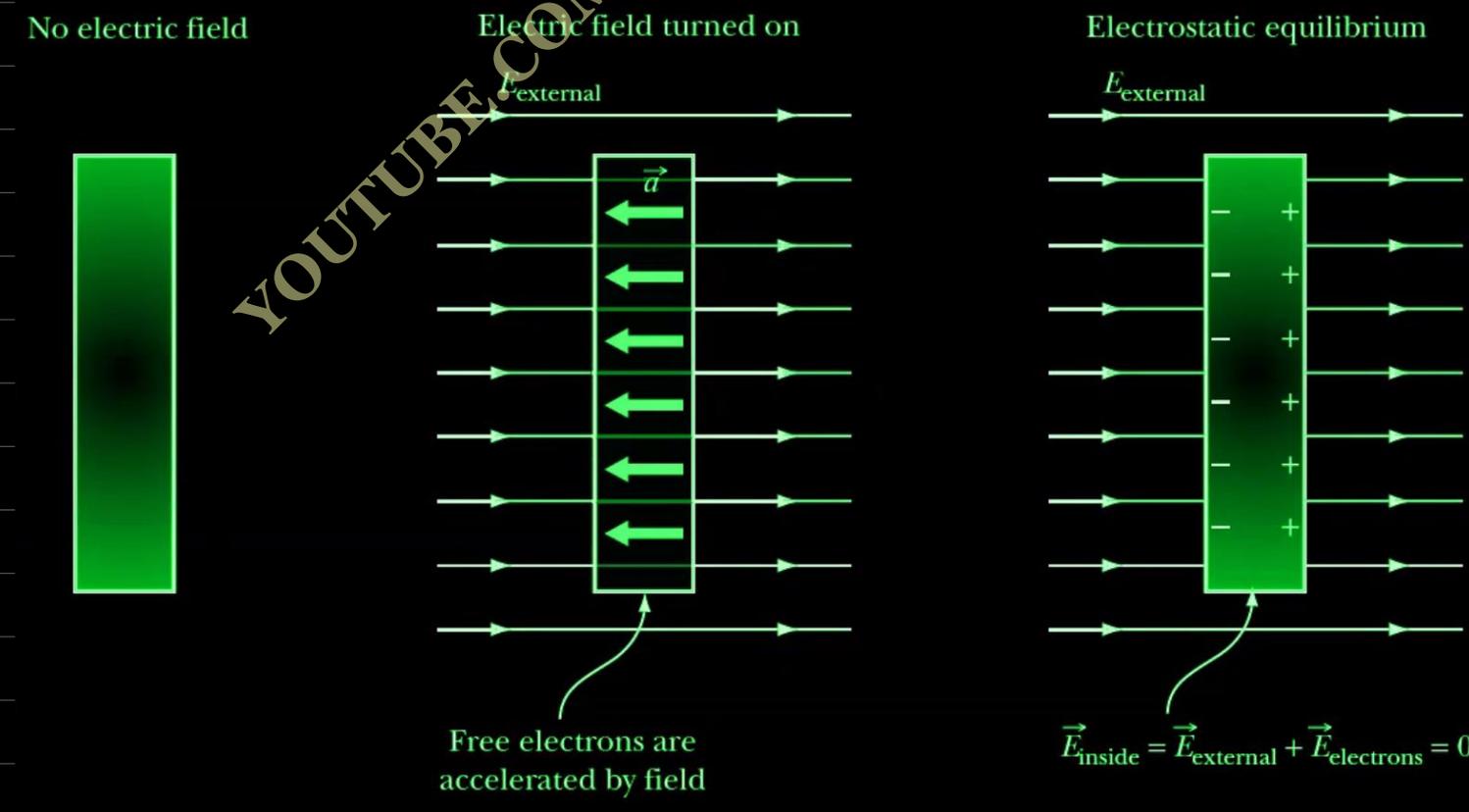
CONDUCTOR : Materials in which charges can freely move from one point to another (macroscopic motion is possible and not just microscopic) are called conductors. For instance metals are conductors because they have free electrons which can freely move within the matrix of the metal. The outermost shell electrons don't stay bonded to their own nucleus but depending on applied fields they can travel. In the metals electrons are as free as fish in the oceans, they are free to move within the metal matrix but not free enough to break free from the metal crystal under normal circumstances. While in non conductors, electrons can't make macroscopic movements within the substance. Only small movements wrt respective nuclei are possible. Apart from metals some non metals can also permit free flow of electrons due to their crystal structures (being governed by laws of quantum mechanics), for example, graphite is a fairly good conductor despite being a non-metal.

PROPERTIES OF CONDUCTORS:

1. ZERO FIELD PROPERTY:



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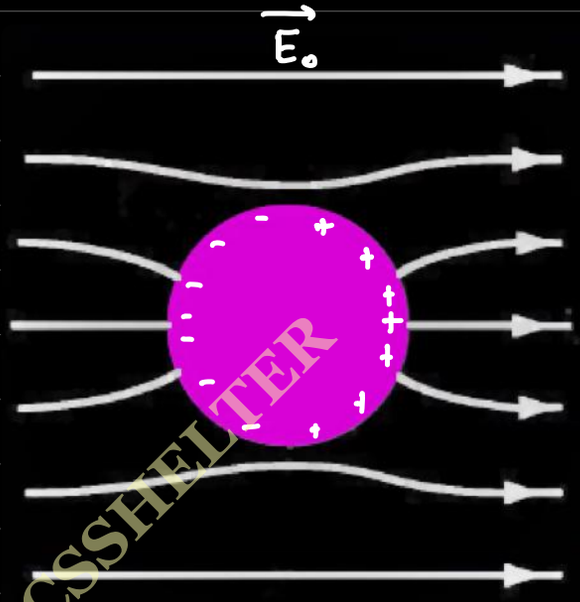
In steady state, net electric field in a conductor is always zero, because until the field in the matrix of a conductor becomes zero, the electrons will keep on moving against the field and hence keep diminishing the field.

EXAMPLE: A metallic sphere is placed in a uniform external electric field $\vec{E}_0$. What is the field due to induced charges inside the sphere. Can you guess the surface charge distribution to generate this field?

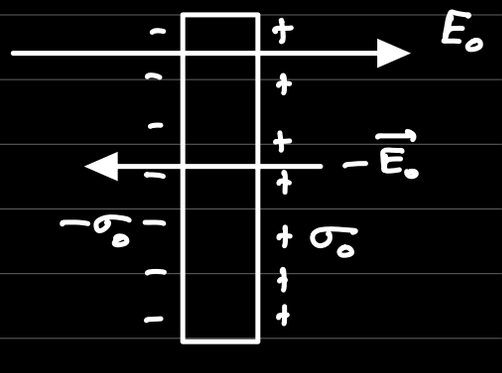


Solution: ∵ Net field inside is zero
∴ Field due to induced charge must exactly cancel the external field
∴ $\vec{E}_{ind} = -\vec{E}_0$. Now we have studied that a surface charge density function of the form $\sigma_0 \cos\theta$ generates a uniform field inside of magnitude $\sigma_0/3\epsilon_0$. Thus to generate a cancelling field our guess would be be a surface charge density of the form $\sigma = \sigma_0 \cos\theta$ such that $\frac{\sigma_0}{3\epsilon_0} = E_0$ or $\sigma_0 = 3\epsilon_0 E_0$.

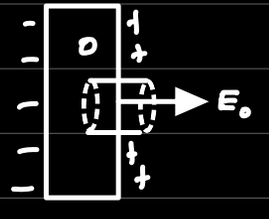
→ Recall that for outside points this behaves like a point dipole of dipole moment $\frac{4}{3}\pi R^3 \sigma_0$.

→ In case of a long cylinder also, a charge density of $\sigma_0 \cos\theta$ would appear with $\sigma_0 = 2\epsilon_0 E_0$

→ With an infinite sheet uniform charge densities of $+\sigma_0$ and $-\sigma_0$ will appear such that $\frac{\sigma_0}{\epsilon_0} = E_0$ or $\sigma_0 = \epsilon_0 E_0$.



This can be readily seen by Gauss Law
$E_0 \times S + 0 \times S = \sigma_0 \times S/\epsilon_0 \Rightarrow \sigma_0 = \epsilon_0 E_0$.



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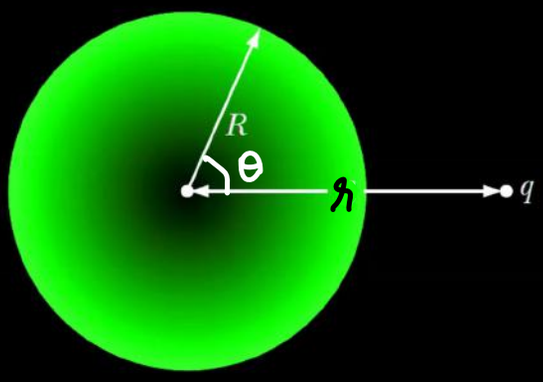
EXAMPLE: A neutral conducting sphere is placed in the field of a point charge as shown in the figure. Find the $|\vec{E}|$ due to induced charges just below the surface at a point defined by Variables (R, θ).

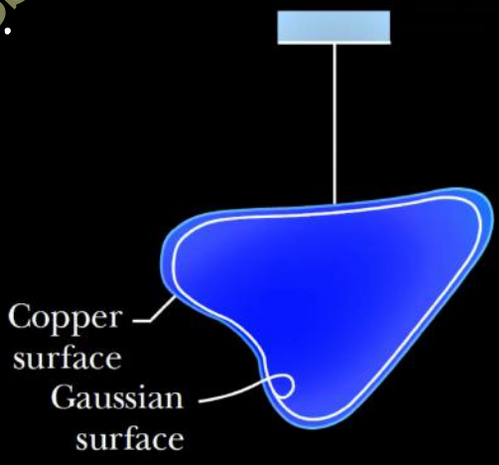


Solution: Just below the surface, net field is zero ∴ it must exactly cancel the field due to q.
Thus required E:

$$|\vec{E}| = \frac{kq}{R^2 + r^2 - 2Rr\cos\theta}$$ Ans.

2. ALL CHARGE GIVEN TO A CONDUCTOR RESIDES ON ITS SURFACE:

This property directly follows from zero field property. ∵ net field inside the conductor is zero ∴ flux through any arbitrary Gaussian surface completely in the metal matrix is zero and therefore net charge enclosed in any such surface is zero. ∴ We can say that whatever extra charge is given to a conductor cannot reside in its matrix and ∴ it must reside on its surface. (In case of conductor with cavities, this charge may also be present on cavity surfaces though not necessarily).



→ This statement is based on a macroscopic average of charges and fields and not microscopic fields. When we say field in a conductor is zero, we obviously don't mean absence of field between the nuclei and surrounding electrons. Our statement is based on a continuum model of conductors.

3. ENTIRE VOLUME OF A CONDUCTOR IS EQUIPOTENTIAL VOLUME:

Since net field in the matrix of a conductor is zero, ∴ no work done in going from one point in the matrix of the conductor to any other point. Recall the potential difference is negative

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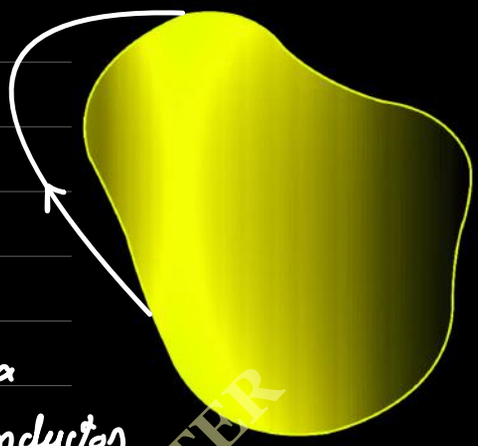
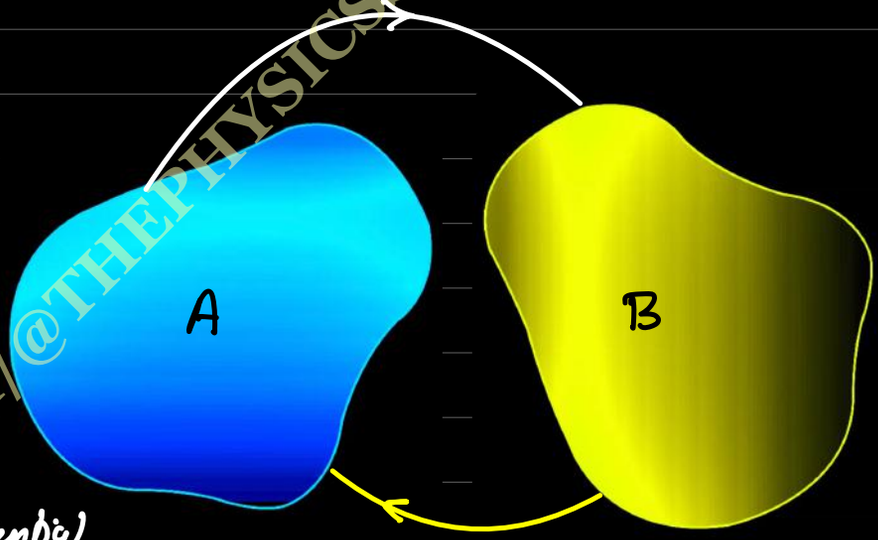
of work done by electrostatic force, thus entire conductor volume must be an equipotential volume.

EXAMPLE: Can a field line originating from a conductor terminate on the same conductor as shown in the figure?



Solution: As we move along a field line, the potential always decreases, however a conductor is equipotential everywhere, thus a field line to itself would mean that conductor is at a lower potential than itself which is absurd. ∴ it is not possible.

EXAMPLE: A and B are two conductors whose two field lines are shown. Is this type of field line pattern possible?



Solution: Upper field line implies that B is at lower potential than A, however lower field line implies that A is at lower potential than B, this is an obvious contradiction. ∴ This type of field lines are not possible.

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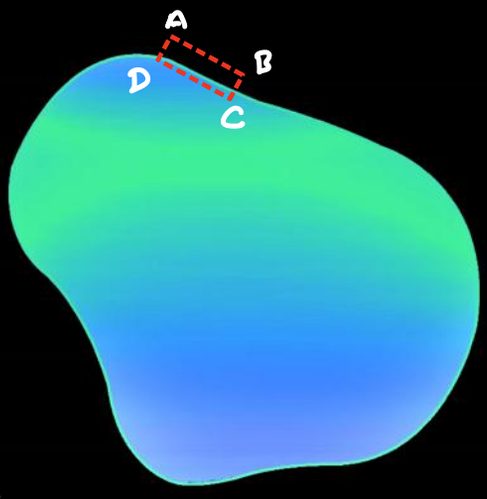
4. NEAR THE SURFACE OF A CONDUCTOR FIELD LINES ARE ⊥ TO SURFACE

Consider a closed path ABCD very near to the surface of the conductor with sides AD and BC negligibly small. The net work done in this closed path must be zero. However since field at CD = 0 (∵ Conductor matrix) ∴ Work done along AB must also be zero. This is possible if and only if tangential



field at AB is zero. ∴ field lines must be ⊥ to surface of the conductor.

5. Field Near Surface Of Conductor Is σ/ϵ_0 :

Consider a pill shaped Gaussian surface near the surface of a conductor. Applying Gauss law:

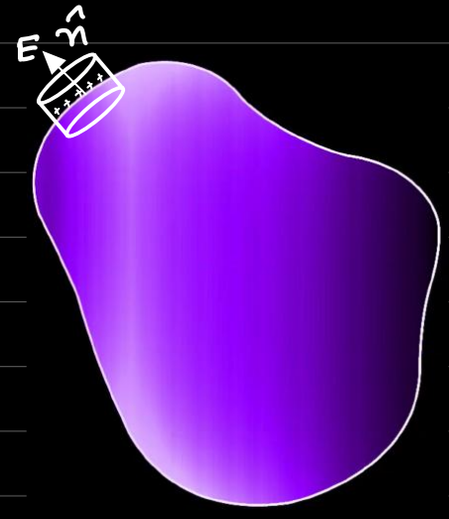
$$ES = \sigma S/\epsilon_0$$

$$\Rightarrow \boxed{E = \frac{\sigma}{\epsilon_0}}$$

→ Here σ is local charge density.

→ Here E is due to all charges in the universe not just due to conductor alone (Though σ will depend on configuration of the universe).

6. Electrostatic Pressure At Surface Of Conductor Is $\sigma^2/2\epsilon_0$ $(= \frac{1}{2}\epsilon_0 E^2)$



Field near surface: $\frac{\sigma}{\epsilon_0}\hat{n}$ where $\hat{n}$ is normal to surface.

Let $\vec{E}_1$ be contribution of the local patch and let $\vec{E}_2$ be the contribution due to rest of the universe. Now for just outside Conductor:

$$\vec{E}_1 + \vec{E}_2 = \frac{\sigma}{\epsilon_0}\hat{n} \quad (1)$$

Just Inside:

$$-\vec{E}_1 + \vec{E}_2 = 0 \quad (2)$$

From (1) & (2), $\vec{E}_1 = \vec{E}_2 = \frac{\sigma}{2\epsilon_0}\hat{n}$. Now let dS be the local patch area, then:

$$d\vec{F} = \sigma\, dS \times \vec{E}_2 = \frac{\sigma^2}{2\epsilon_0} dS\, \hat{n}$$

Now Pressure $P = \left(\frac{d\vec{F}}{dS}\right) = \frac{\sigma^2}{2\epsilon_0} \Rightarrow \boxed{P = \frac{\sigma^2}{2\epsilon_0}}$

Also $u = \frac{1}{2}\epsilon_0 E^2 = \frac{1}{2}\epsilon_0\left(\frac{\sigma}{\epsilon_0}\right)^2 = \frac{\sigma^2}{2\epsilon_0} = P.$

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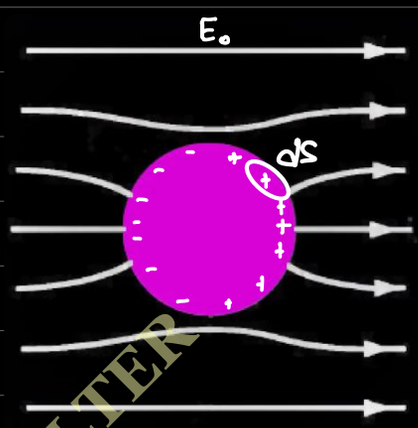
EXAMPLE: When a solid conducting sphere of radius R is placed in a uniform electric field E_0, the electric force ripping apart its two hemispheres is F, by what factor does this force increase if we take a sphere of radius ηR and increase the field to nE_0.



Solution: Recall that charge distribution is of the form $\sigma_0 \cos\theta$ with $\sigma_0 = 3\epsilon_0 E_0$, Now consider a patch of area dS at an angle θ, Electrostatic force on this $dF = PdS$

$$\Rightarrow dF = \frac{(3\epsilon_0 E_0 \cos\theta)^2}{2\epsilon_0} dS$$

$\Rightarrow dF \propto E_0^2 dS$, Now when radius of sphere is made η times and Electric field is made n times, surface area becomes η^2 times, thus force becomes $n^2\eta^2 F$.

7. **EXISTENCE THEOREM:** For a given charge configuration of the universe and a given total charge on the conductor, there always exists a charge configuration of the conductor for which field inside the conductor can be made zero throughout (or the conductor can become equipotential). The theorem is not explicitly within JEE syllabus but plays an important role in understanding the important property of electrostatic shielding hence we have stated here (without proof). (The proof involves an understanding of theory of partial differential equations).

8. **UNIQUENESS THEOREM:** For a given total charge on a conductor and a charge configuration of the universe, the charge configuration of the conductor for which field in the conductor becomes zero will be unique (if exists).

→ Existence theorem says that there exists atleast one configuration of the conductor for which field inside can become zero, while uniqueness theorem says that there exists at most one such configuration, combining these two theorems we can say that there exists one

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and only one configuration of the conductor (for given fixed config. of universe and total charge on the conductor) for which the field inside the conductor will become zero everywhere.

→ This has a profound consequence for it means that if somehow we are able to find such a configuration (may be through an intelligent guess) then we need to look no further for any other solution, for there is no other solution except the one that we guessed.

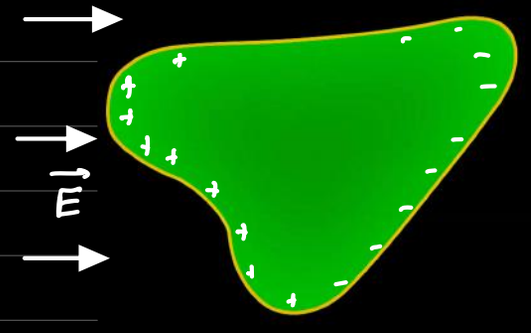
9. Field In "Unoccupied" Cavity Of A Conductor Is Zero:

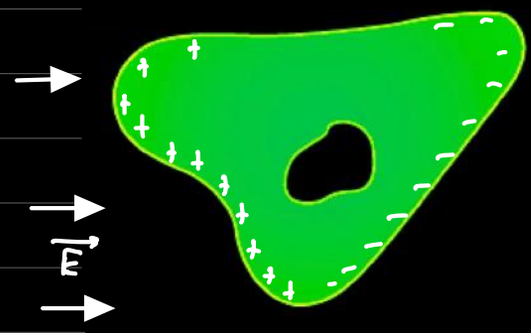
We shall provide two proofs of this fact.

(I) Suppose field lines exist in an occupied cavity, they must emanate from surface of cavity and also terminate at the cavity surface, thus making the conductor at lower potential than itself. (Alternatively if we move along such a field line and complete the circuit through matrix of conductor, then clearly work done in such a closed path will be non-zero. (Note that for such field lines to exist, cavity will have to get equal and opposite charges as can be readily seen by drawing a Gaussian Surface just outside the cavity.

(II) By Using Existence and Uniqueness Theorem:

Consider first the conductor without the cavity, If it has some net charge, it will distribute it on its outer surface such that superposition of universe's field with conductor's own field gives a zero net field. Now if we create a cavity in such a conductor by digging up neutral material, the field in the volume still remains zero (including cavity region). Since we have found one way of making field in the conductor, this must be





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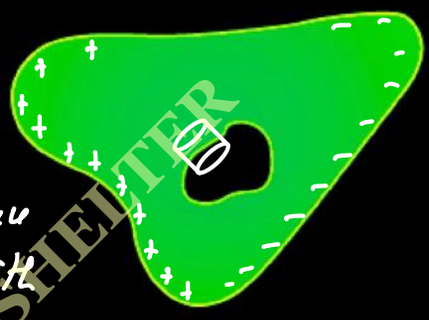
the only configuration, and it does make the field in cavity also zero.

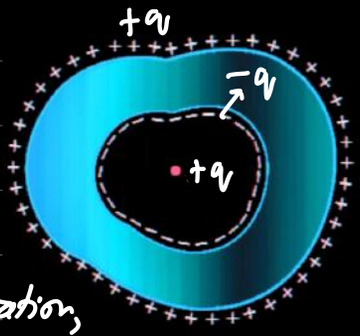
10. Surface Charge Density Of "Unoccupied" Cavity Is Zero:

Since there exists no field lines in an unoccupied cavity, there cannot be any local charge density on cavity surface.
(Because from a positive charge, field lines must emanate and terminate on a negative charge). Alternatively we can easily show that such a charge density on an unoccupied cavity surface will violate Gauss' law. (Consider the Gaussian surface shown in the figure. Net flux associated with it is zero ∵ neither there is field in cavity nor in the metal matrix ∴ it can't have any enclosed charge ∴ $\sigma = 0$ on surface of cavity).



11. Surface Of "Occupied" Cavity Gets Negative Of The Charge Enclosed:

Suppose we place a charge $+q$ in the cavity of a neutral conductor as shown in the figure. Then due to induction, cavity surface will get negatively charged. If we make a Gaussian surface through metal matrix, tightly hugging the cavity everywhere, then flux through this surface must be zero ($\because E = 0$ in metal matrix) ∴ cavity surface must have exactly negative of the charge enclosed so that net charge enclosed by our Gaussian surface may be zero. Now since conductor is overall neutral, for charge conservation, a $+q$ will move towards outer surface of the conductor.



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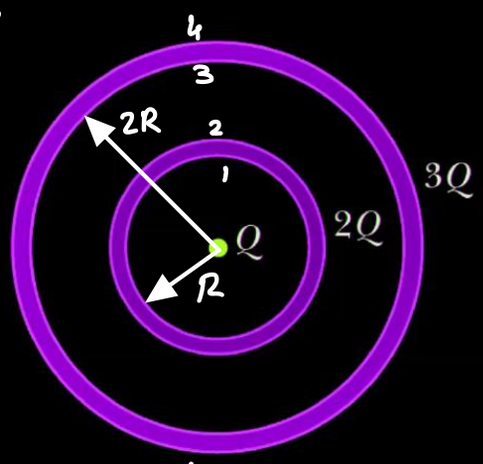
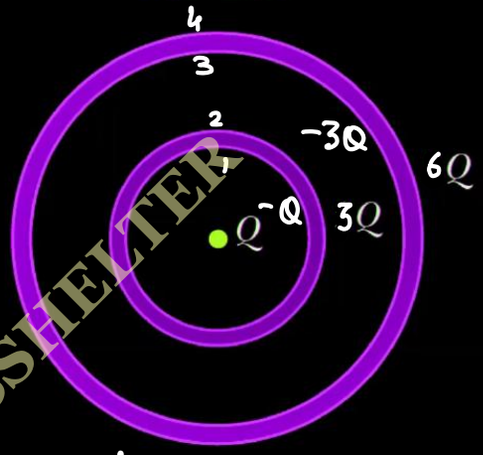
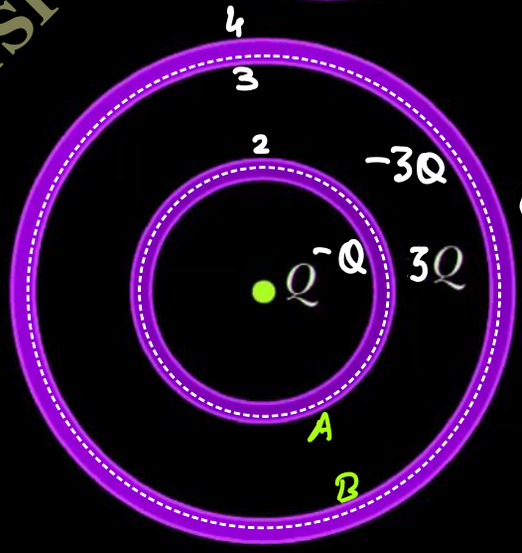
EXAMPLE: A point charge $+Q$ is enclosed inside two thin spherical shells which in turn have net charges $2Q$ and $3Q$ respectively. Find net charge on each surface 1, 2, 3 and 4. Also find potentials of inner and outer shells.



Solution: For inner shell, Q is is the enclosed charge $\therefore$ surface ① gets $-Q$ and by charge conservation ② gets $+3Q$. For outer shell, $3Q$ is the net enclosed charge in its cavity, thus surface ③ gets $-3Q$ and by charge conservation outer surface ④ must get $+Q$.



→ Note that in a system of shells enclosing other shells (not necessarily concentric) [shells may be spherical or cylindrical] Facing faces will always get equal and opposite charges, Because any gaussian surface passing through metal must have zero flux (Hence zero total enclosed charge) [For clear understanding focus on Gaussian surfaces A & B, Both have net zero charge enclosed $\therefore$ 2 & 3 have equal and opposite charges.



Now Potential at inner shell: $V_{in} = \underbrace{k \times \frac{3Q}{2R}}_{\text{outer shell}} + \underbrace{k \times \frac{2Q}{R}}_{\text{Inner shell}} + \underbrace{k \times \frac{Q}{R}}_{\text{Point charge.}} = \frac{9kQ}{2R}$

→ Note that for an inside point potential contribution of a shell is same as that its surface.

→ Since shells are thin we just need to consider net charges on shells for writing potentials. However if the shells were thick, we will need to consider individual charges on each surface for calculations potentials

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Now $V_{out} = k \times \frac{3Q}{2R} + k\frac{2Q}{2R} + \frac{kQ}{2R} = \frac{3kQ}{R}$ Ans.

(Here all denominators are $2R$, due to outer shell surface being an outside point).

EXAMPLE IRODOV 3.140:

A point charge q is located at the centre O of a spherical uncharged conducting layer provided with a small orifice (Fig. 3.32). The inside and outside radii of the layer are equal to a and b respectively. What amount of work has to be performed to slowly transfer the charge q from the point O through the orifice and into infinity?

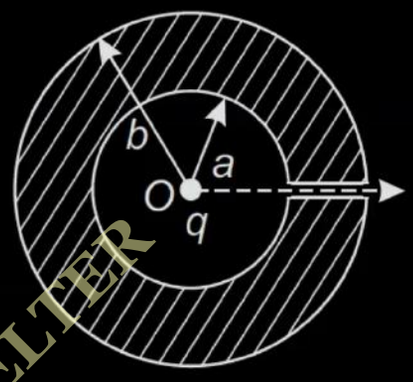


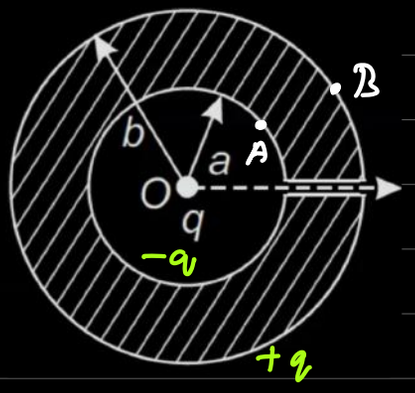
Fig. 3.32

Solution: METHOD I: SELF ENERGY & INTERACTION ENERGY APPROACH

For purpose of calculating electrostatic Energy, we have to consider following energies initially:

Self Energy: A & B, Interactions: A-B, A-O, B-O

Thus $U_i = \underbrace{kq^2/2a}_{A} + \underbrace{kq^2/2b}_{B} + \underbrace{\frac{-kq^2}{a}}_{OA} + \underbrace{\frac{kq^2}{b}}_{OB} - \underbrace{\frac{kq^2}{b}}_{AB}$



$\Rightarrow U_i = \frac{kq^2}{2}\left[\frac{1}{b} - \frac{1}{a}\right]$, finally all of them become zero when O is removed to ∞. Thus $U_f = 0 \Rightarrow \Delta W = U_f - U_i = \frac{kq^2}{2}\left[\frac{1}{a} - \frac{1}{b}\right]$

METHOD II: CONSIDERING ENERGY CHANGES IN SPACE: Initially a thick shell like region between radii a & b around the point charge is devoid of electrostatic energy, however finally this region around the charge also stores the energy and remaining space still contains the same amount of energy (As seen from frame of point charge O). Thus amount of work done must be equal to electrostatic energy stored in the thick shell like volume between radii a & b, which is nothing but difference in self energy of shells of radii a and b i.e $\frac{kq^2}{2a} - \frac{kq^2}{2b} = \frac{kq^2}{2}\left(\frac{1}{a} - \frac{1}{b}\right)$ Ans.

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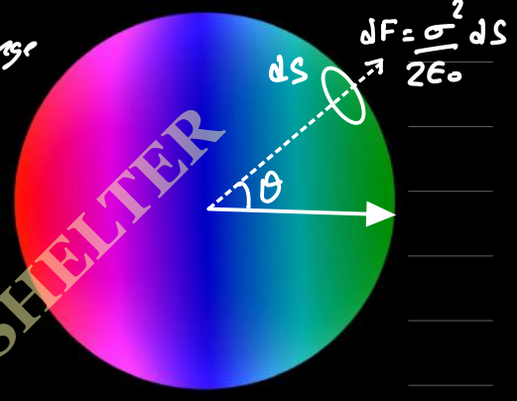
Example : Krotov 3.7

The minimum strength of a uniform electric field which can tear a conducting uncharged thin-walled sphere into two parts is known to be E_0. Determine the minimum electric field strength E_1 required to tear the sphere of twice as large radius if the thickness of its walls is the same as in the former case.

Solution: Since all charge given to a conductor resides on its surface and the unoccupied cavity has a zero field, essentially the behaviour of a conducting shell with unoccupied cavity is same as that of a solid conducting sphere. Thus to create a zero field within, shell acquires a surface charge configuration $\sigma = \sigma_0 \cos\theta$ such that $E_0 = \sigma_0/3\epsilon_0$



Now if we have a patch of area dS on the shell,

electrostatic force on the patch $= dF = \frac{\sigma^2}{2\epsilon_0} dS$

$\therefore dF \propto E_0^2 \times dS \Rightarrow F \propto E_0^2 \times R^2$, Now let breaking stress be σ_b and wall thickness be t, then $\sigma_b \times 2\pi R \times t = C E_0^2 R^2$ (1), Now let breaking field for shell of double radius be ηE_0, then similar to equation (1) we can also write: $\sigma_b \times 2\pi (2R) t = C(\eta E_0)^2 \times (2R)^2$ (2)

(2) / (1) $\Rightarrow$

$$2 = 4\eta^2 \Rightarrow \eta = \frac{1}{\sqrt{2}}$$

$\Rightarrow$ Required field $E' = E_0/\sqrt{2}$ Ans.

12. Electrostatic Shielding:

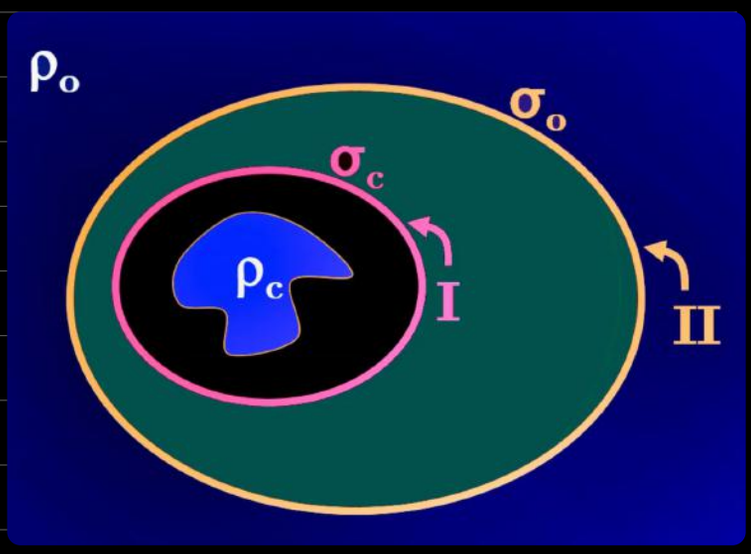
Symbol definitions:

$\rho_c \rightarrow$ Charge density function in cavity of conductor.

$\sigma_c \rightarrow$ Charge density function on cavity surface.

$\rho_0 \rightarrow$ Charge density function in outer universe

$\sigma_0 \rightarrow$ Charge Density Function on outer surface of conductor.



Now Electrostatic shielding can be stated in symbols as:

(1) $\vec{E}_{\rho_c} + \vec{E}_{\sigma_c} = 0$ Everywhere outside boundary I

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② $\vec{E}_{\rho_0} + \vec{E}_{\sigma_0} = 0$, Everywhere inside boundary II.

Or Casually we can say that inner world creates zero field in the middle world and outer world and outer world creates zero field in the middle world and inner world. Thus inner world is as good as non-existent for middle and outer world and outer world is as good as non existent for middle and inner worlds as far as electric fields are concerned.

Proof of Electrostatic Shielding:

Claim 1: For any given volumetric charge distribution ρ_c and a given surrounding surface, there always exists a surface distribution function σ_c such that the combined field of ρ_c and σ_c is zero everywhere outside the surrounding surface.

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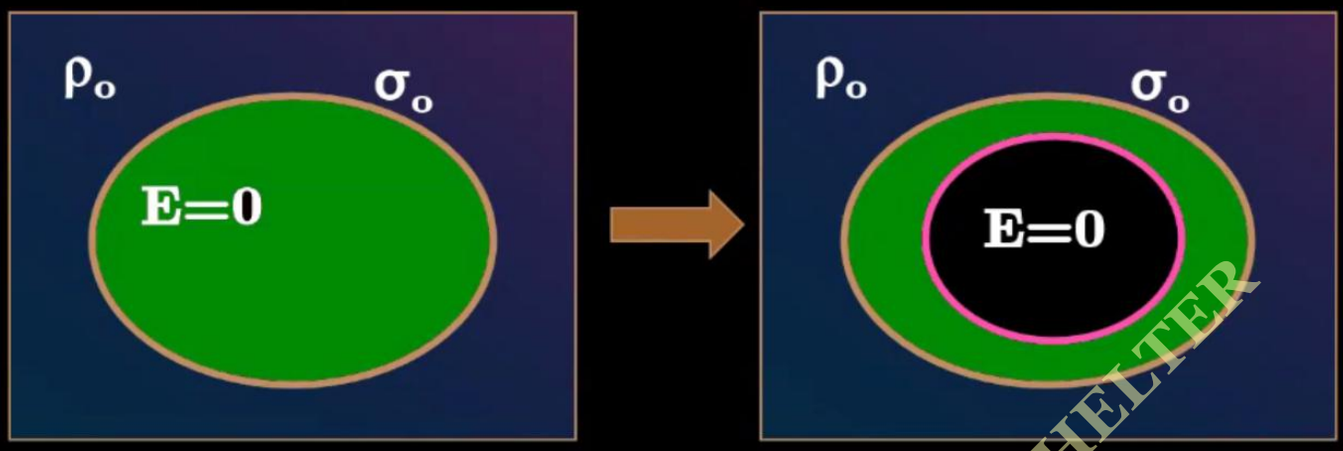
First we imagine an infinite conductor with cavity, let ρ_c be the charge density function in the cavity, then by existence and uniqueness theorem, there will be some charge density function σ_c, such that field due to ρ_c and σ_c is zero all the way up to infinity. Now if we start digging up neutral material from infinity all the way up to a finite point, the field still remains zero. Thus there always exists a way to create zero field in any finite conductor with a cavity just using the cavity volume charges and cavity surface charges.

→ CONCLUSION I: FOR EVERY ρ_c THERE EXISTS A UNIQUE σ_c SUCH THAT ρ_c & σ_c CREATE A ZERO FIELD EVERYWHERE OUTSIDE THE CAVITY.

Claim 2: For any given volumetric charge distribution ρ_0 in the universe and a given conductor with some given charge, there exists a surface distribution σ_0 such that net field in the conductor becomes zero due to the charge of the universe and the charge on the conductor surface.



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Consider a solid conductor without any cavity. If ρ_0 is the charge density function of the universe, let σ_0 be the charge density function on outer surface of the conductor such that net field in the conductor due to ρ_0 and σ_0 becomes zero. Now if we dig up neutral material from within the conductor to form a cavity, σ_0 should not change because σ_0 & ρ_0 still maintain a zero field in the conductor matrix and by existence and uniqueness theorem, this must be the only way to accomplish the zero field. Thus we have conclusion II:

CONCLUSION II: FOR ANY CHARGE DISTRIBUTION IN THE UNIVERSE, WE CAN ALWAYS GENERATE ZERO FIELD IN THE CONDUCTOR (INCLUDING ONE WITH A CAVITY) BY A UNIQUE SURFACE CHARGE DENSITY σ_0 ON OUTER SURFACE OF THE CONDUCTOR.

FROM CONCLUSION I & II, we have a way of creating a zero field in a conductor such that cavity volume and cavity surface together produce a zero field in the conductor volume and Universe and outer surface of conductor together produce a zero field in the conductor volume. ∴ By uniqueness and existence theorem this must be the only way of creating zero field in the

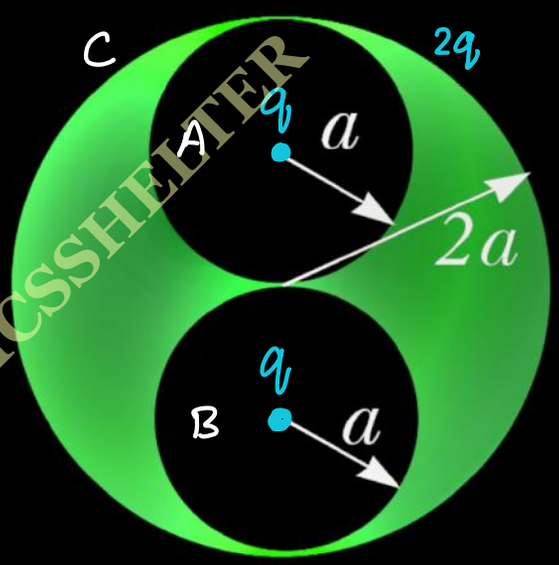
conductor. Thus $\vec{E}_{p_c} + \vec{E}_{\sigma_c} = 0$ Everywhere outside the cavity

and $\vec{E}_{p_0} + \vec{E}_{\sigma_0} = 0$ Everywhere in the volume of the conductor

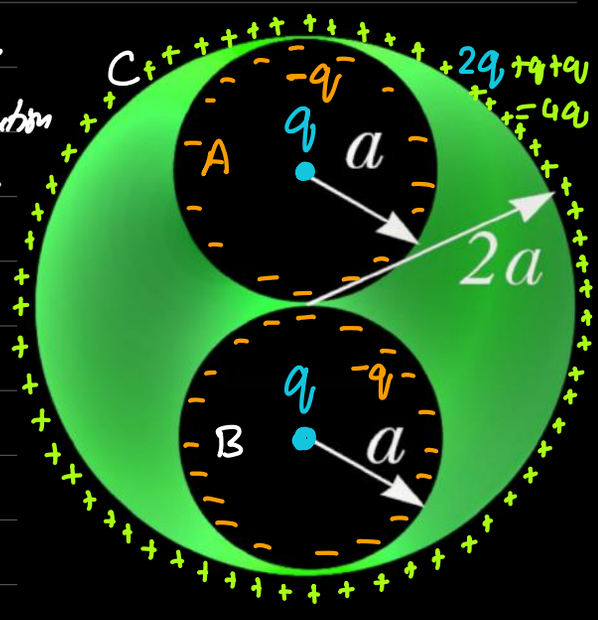
and its cavities.

→ Same principle also applies to conductors with multiple cavities as well. i.e. summation of field due to cavity surface and cavity volume charges is zero everywhere outside that cavity.

EXAMPLE: A conductor C has a net charge 2q. Now two charges q each are placed at centres of its two cavities as shown. Find (a) Charges appearing on each surface (also comment on its distribution).
(b) Field and potential in cavities as a function of distance r from their centres
(c) Field and potential outside the conductor as a function of distance r from it centre.



Solution: (a) Each cavity is shielded from middle and outer world, thus while considering distribution in the cavity we can treat everything outside it as good as non existent. Also the cavity surface and volume charges must produce zero field everywhere outside the cavity, this is achieved by having a uniformly spread negative charges on cavity surface. Thus both cavities have uniformly spread -q each on their surfaces as shown. Now by charge conservation, a +q must move out to the outer surface of the conductor to compensate for a -q appearing on cavity surface of each cavity. Thus outer surface gets a total charge of 2q+q+q = 4q.



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Now for purpose of distribution of this charge, cavities are as good as non existent due to shielding property (∵ cavity surface and volume charges produce a zero field everywhere outside the cavity). Thus this $+4q$ gets uniformly distributed on outer surface of the conductor.

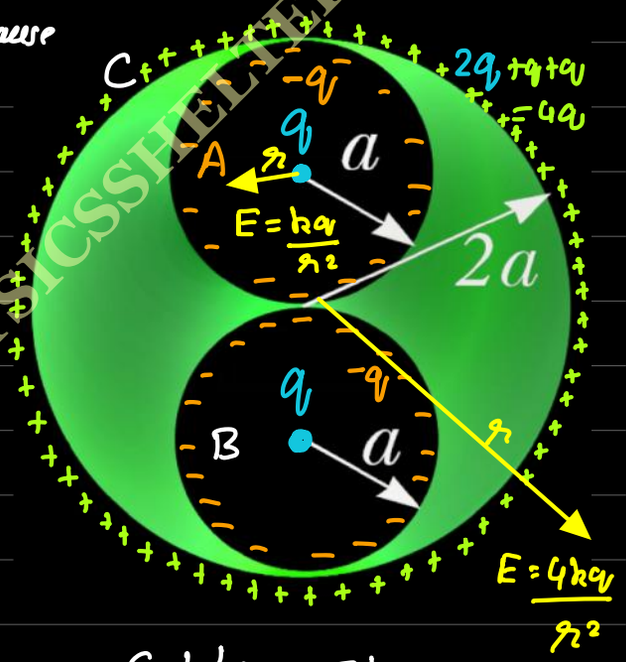
(b) Fields inside the cavity surfaces must be kq/r^2 in both cavities because cavity surface charges won't produce any field due to shell theorem. (r is the distance from cavity centres).

Now potential of cavity surfaces must be same as the potential of the outer surface because entire conductor is an equipotential volume.

Now potential of outer surface :

$$V_s = \frac{k \times 4q}{2a} = \frac{2kq}{a}$$

(∵ Cavities do not matter in computing work from ∞ to outer surface of conductor).



Now inside Cavity : $E = \frac{kq}{r^2} = -\frac{dV}{dr}$

$\Rightarrow V = \frac{kq}{r} + C$, also $V(a) = V_s = \frac{2kq}{a} \Rightarrow C + \frac{kq}{a} = \frac{2kq}{a}$

$\Rightarrow C = \frac{kq}{a} \Rightarrow V = \frac{kq}{r} + \frac{kq}{a}$ Ans.

(c) For outside points, cavities are as good as non-existent, thus outside the conductor:

$$V = \frac{k \times 4q}{r} = \frac{4kq}{r} \text{ Ans.}$$

$$E = \frac{k \times 4q}{r^2} = \frac{4kq}{r^2} \text{ Ans.}$$

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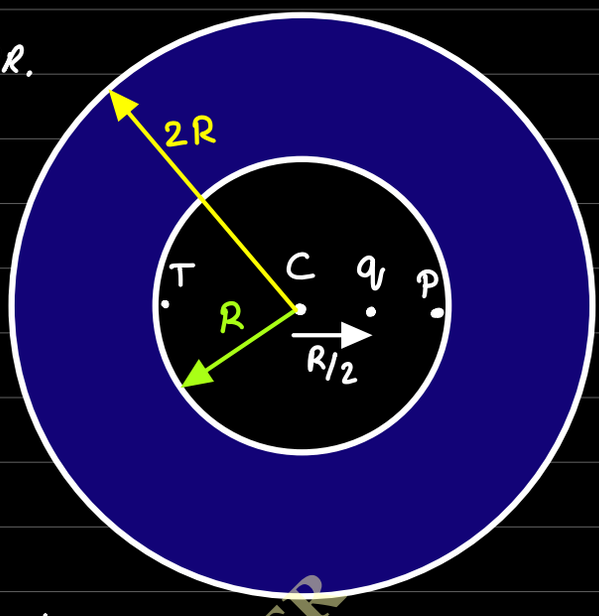
EXAMPLE: Consider a thick neutral conducting shell with inner radius R and outer radius 2R. A point charge q is kept at a distance R/2 from the centre. P and T are points just inside the cavity surface, collinear with centre C and the point charge.



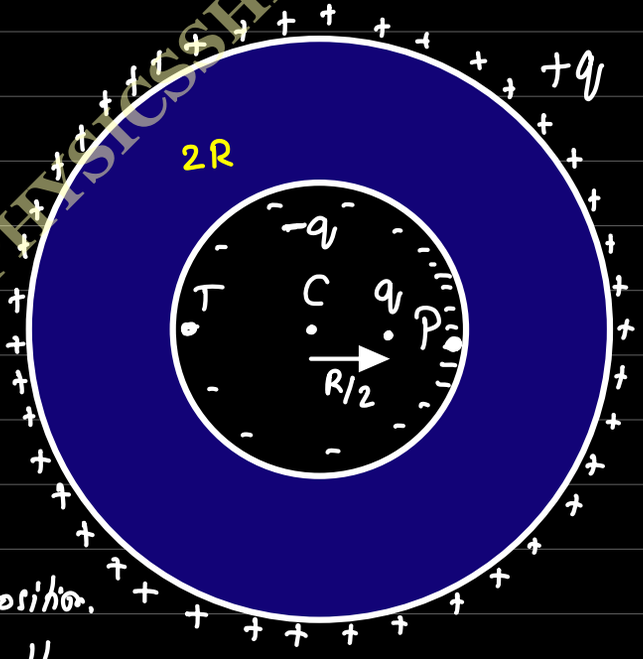
Find: (a) Potential at point P

(b) Potential at point C

(c) Contribution of cavity surface charges to potential of P

(d) Find field due to cavity surface charges at a point in metal matrix very close to point T.

Solution: Consider the charge distribution as shown in the adjacent figure. (Note that charge on outer surface is uniformly distributed). Now ∵ metal is equipotential ∴ potential at P must be same as potential of outer surface. Thus:



(a) $V_P = kq/2R$. Ans.

(b) Potential at C can be found by superposition. Note that cavity surface charges are all equidistant from C, thus their contribution to potential of C is simply $-kq/R$. Now by superposition:

$$V_C = \underbrace{\frac{kq}{R/2}}_{\text{point}} - \underbrace{\frac{kq}{R}}_{\text{cavity}} + \underbrace{\frac{kq}{2R}}_{\text{outer}} = \frac{3}{2}\frac{kQ}{R} \text{ Ans.}$$

(c) Total potential at P is $kq/2R$. Now to find contribution of cavity surface charges we can subtract the contributions of point charge and outer surface charges thus:

$$V_{cavity} = \frac{kq}{2R} - \frac{kq}{2R} - \frac{kq}{(R/2)} = -\frac{2kq}{R} \text{ Ans.}$$

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Alternatively we could say that the cavity volume charges and cavity surface charges create a net zero potential everywhere at cavity surface and beyond. ∴ potential contribution of cavity surface charge must be −ve of contribution due to point charge.

Thus $V_{cavity} = -(kq/(R/2)) = -2kq/R$ Ans.

(d) In metal matrix near point T, field due to cavity surface must exactly cancel the field due to the point charge.

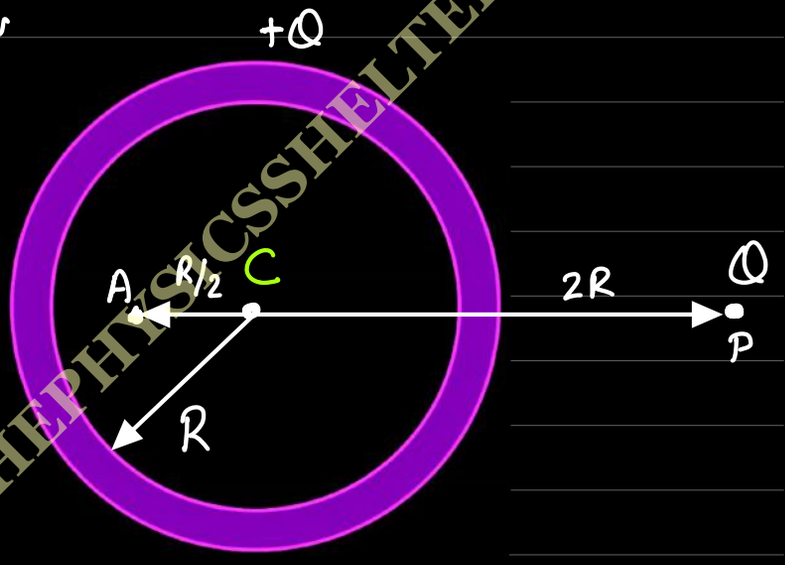
Thus $\vec{E}_{cavity} = \frac{kq}{(3R/2)^2}\hat{i} = \frac{4}{9}\frac{kq}{R^2}\hat{i}$ Ans.

EXAMPLE: A thin conducting shell has a total charge Q. Another point charge Q is placed outside it as shown in the figure. Find

(a) Potential at A due to point charge at P

(b) Total potential at point A

(c) Potential at A due to surface charges.



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Solution: (a) Potential at A due to point charge Q, $V_P = \frac{kQ}{2.5R} = \frac{2}{5}\frac{kQ}{R}$

(b) Potential at A must be same as potential at C ∵ field in the cavity is zero (∵ no work is done in going from C to A).

$\therefore V_A = V_C = \frac{kQ}{R} + \frac{kQ}{2R} = \frac{3kQ}{2R}$ (∵ Surface charges are equidistant from C).

(c) Potential contribution at A due to surface:

$V_{cavity} = \frac{3kQ}{2R} - \frac{2}{5}\frac{kQ}{R} = \frac{11}{10}\frac{kQ}{R}$ Ans.

13. Linear Superposition Of Induced Charges On Neutral Conductors :

Let σ_1 be the induced surface charge function on a neutral conductor corresponding to a surrounding volume distribution ρ_1 and let σ_2 be the surface distribution on the conductor corresponding to a surrounding distribution of ρ_2. Then if surrounding charge

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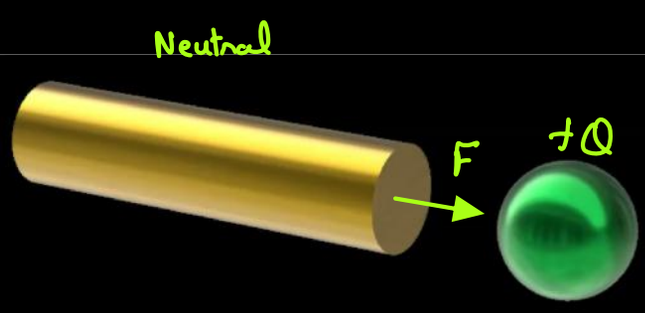
distribution were to be made $\lambda\rho_1 + \mu\rho_2$ then the corresponding surface distribution will be $\lambda\sigma_1 + \mu\sigma_2$.

EXAMPLE: Force of attraction between a charged metallic sphere having a charge Q and a metallic rod is F. Find the force of attraction if instead of Q, the metallic sphere were given a charge $2Q$.



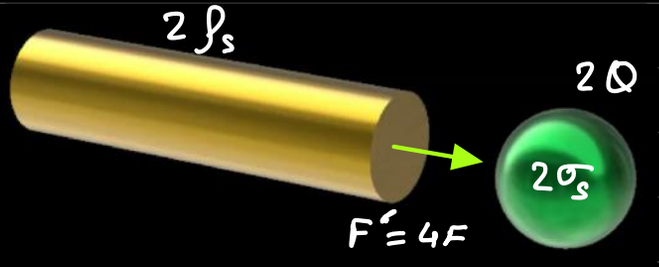
Solution: Let σ_s be the charge distribution function of the sphere and σ_R be the charge distribution function of the rod then σ_s & σ_R together produce a zero field in the rod as well as the sphere. Also for the sphere, $\int \sigma_s dS = Q$. Now when total charge on the sphere is $2Q$, we can easily see that making the charge density functions double on both objects satisfies simultaneously total charges and zero field conditions, thus for the sphere:

$$\sigma_s' = 2\sigma_s \quad \text{and for rod} \quad \sigma_R' = 2\sigma_R.$$

(From uniqueness we know that this is the only possibility.)

Now since every patch has double the initial charge, net force must become four times (∵ force is proportional to product of the two charges).

$$\therefore \quad F' = 4F$$



14. SUPERPOSITION FOR NON-NEUTRAL CONDUCTORS: Let σ_0 be the charge density function on an isolated conductor (Away from all field of the universe), further let σ_1 be the surface distribution on its surface if it were neutral and surrounding density were ρ_1 and let σ_2 be the surface distribution on it if it were neutral and surrounding density were ρ_2. Then, if the conductor were subjected to a surrounding distribution $\lambda\rho_1 + \mu\rho_2$

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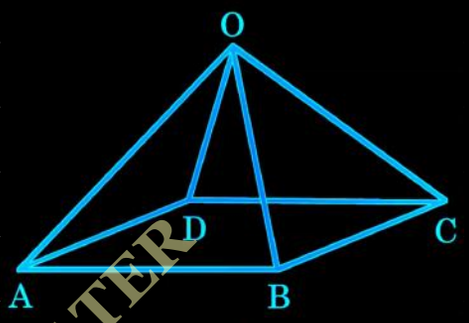
and its total charge were made η times then the new surface distribution is given by:

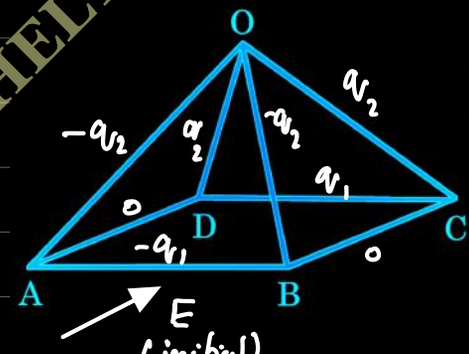
$$\boxed{\sigma = \eta\sigma_0 + \lambda\sigma_1 + \mu\sigma_2}$$

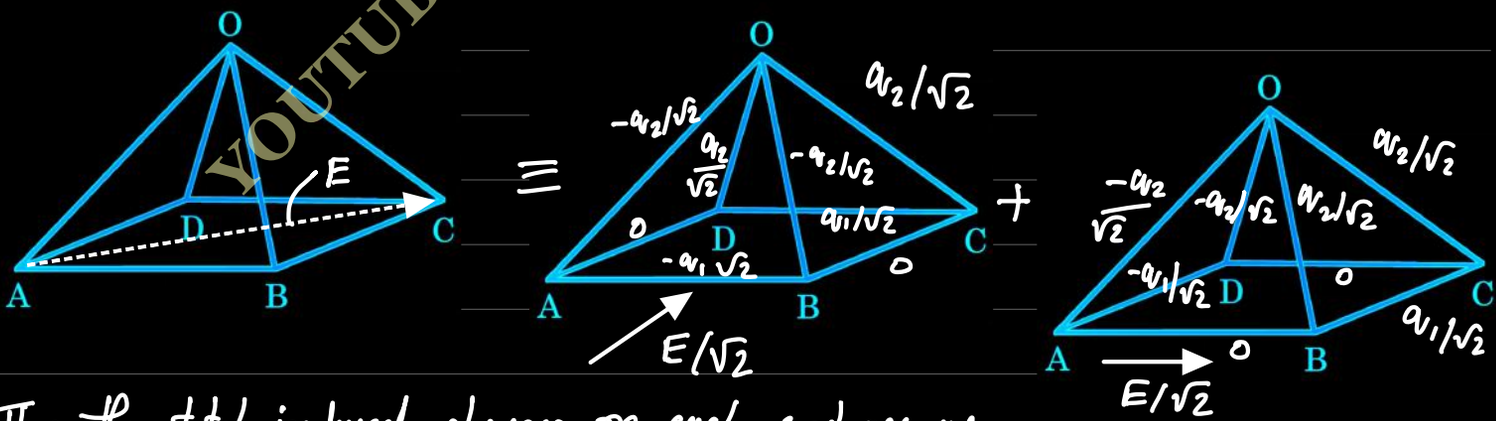
EXAMPLE: PATHFINDER BUILD 4:

A rigid frame in the shape of a right pyramid is made of conducting rods. The base ABCD is a square and the apex O is vertically above the centre of the base. The frame is electrically neutral. When it is placed in a uniform electric field of intensity E pointing from the corner A towards the corner D, total charges induced on the rods DC and OC are known to be q_1 and q_2 respectively. Now the frame is rotated to make the electric field pointing from the corner A towards the corner C. What are the charges induced on each rod?



Solution: The adjacent figure shows the initial distribution of charges. We have used here symmetry in induction, we can always visualise the uniform electric field to be generated by placing the pyramid between two infinite sheets carrying charge densities $+\sigma$ and $-\sigma$. Accordingly if we imagine a vertical plane through O, parallel to side AB, we expect the mirror image points to have equal and opposite charges. Now applying a field E along AC is equivalent to superposition of two fields as shown in figure.





Thus the total induced charges on each rod are as follows:

AB : $-q_1/\sqrt{2}$	AO : $-\sqrt{2}\,q_2$
BC : $q_1/\sqrt{2}$	BO : 0
CD : $q_1/\sqrt{2}$	CO : $\sqrt{2}\,q_2$ Ans.
DA : $-q_1/\sqrt{2}$	DO : 0

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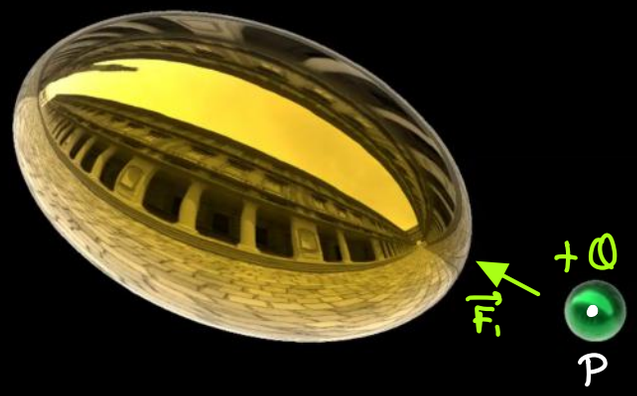
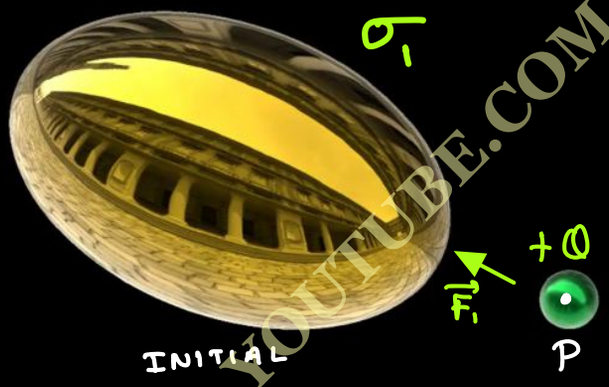
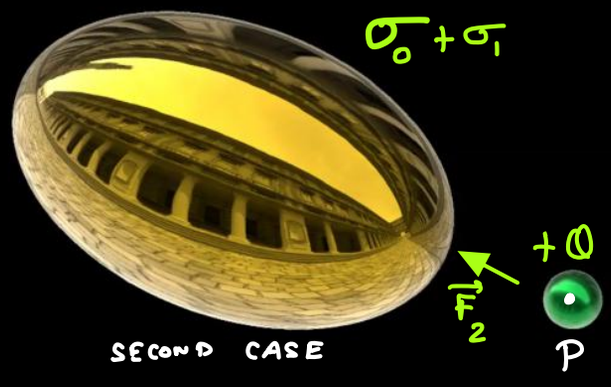
EXAMPLE:

When a charge $+Q$ is placed in front of a neutral conductor, it experiences an electrostatic force $\vec{F}_1$. Now the conductor is given some charge and the force becomes $\vec{F}_2$. What would be the electric field at point P if now the charge $+Q$ were removed to ∞.



Solution: Let σ_1 be the surface charge distribution when the conductor was neutral and in influence of charge $+Q$. Further let σ_0 be the distribution on it when it is given some charge and it is isolated and has some charge (same as the additional charge that was given to it), then by our understanding of superposition, net distribution on it when it is given some charge and is kept in influence of $+Q$, is simply $\sigma = \sigma_0 + \sigma_1$. Now when $+Q$ is removed to ∞, only σ_0 would remain, thus we have following surface distribution and force correspondences:





$$\sigma_1 \longrightarrow \vec{F}_1 = \vec{E}_1 Q \quad (1)$$

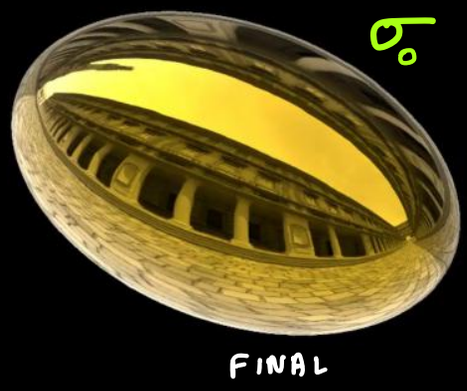
$$\sigma_0 + \sigma_1 \longrightarrow \vec{F}_2 = (\vec{E}_1 + \vec{E}_0) Q \quad (2)$$

From (1) & (2),

$$\vec{E}_0 Q = \vec{F}_2 - \vec{F}_1$$

$$\Rightarrow \vec{E}_0 = (\vec{F}_2 - \vec{F}_1)/Q \quad \text{Ans.}$$

(Here $\vec{E}_0$ is due to σ_0 alone)



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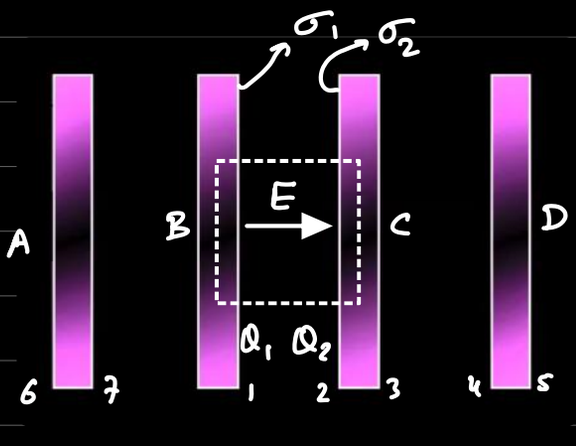
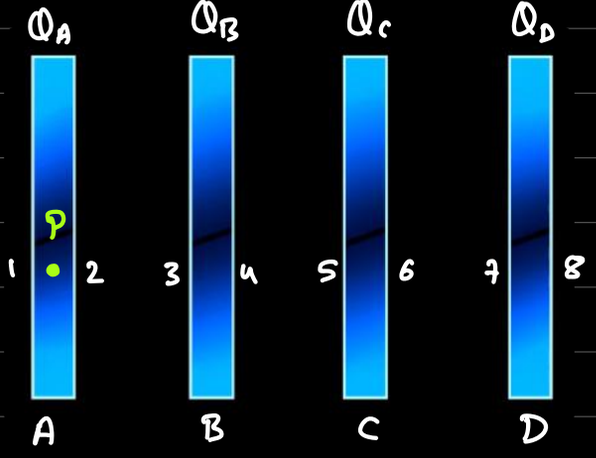
15. FACING FACES OF CHARGED INFINITE (LARGE) SHEETS GET EQUAL AND OPPOSITE CHARGES:



Consider a system of parallel plates, we have drawn a Gaussian cuboid passing through plates B and C as shown in the figure. The plates are very large (Treated like ∞ plates). For this Gaussian Cuboid, Flux through Top, Bottom, Front and back faces must be zero, ∵ field lines are parallel to these faces, also flumes through left and right faces must be zero, ∵ these lie in the metal matrix where net electric field is zero. Thus net flux through Gaussian Cuboid is zero and ∴ net charge enclosed in it must be zero. This can happen only if $\sigma_2 = -\sigma_1$ and if facing areas are equal, $Q_2 = -Q_1$.

It does not matter how many parallel plates are there in the system, such a Gaussian surface can always be drawn between adjacent plates. Thus facing faces in a system of large equal parallel plates will always carry equal and opposite charges.

→ In a system of n parallel plates, extreme faces will get a charge $(\Sigma q_i/2)$ charge each where q_i is the net charge on i^{th} plate. For instance for the shown system of four parallel plates, $Q_1 = Q_8 = \left(\frac{Q_A + Q_B + Q_C + Q_D}{2}\right)$



This can be understood as follows:

① By charge conservation: $Q_1 + Q_8 = Q_A + Q_B + Q_C + Q_D$ ①
(∵ Facing faces add up to zero).

② For net field in metal matrix (say at point P) to be zero, net charge to the right of it must be equal to the net charge to the left of it, ∴ $Q_1 = Q_8$ ②

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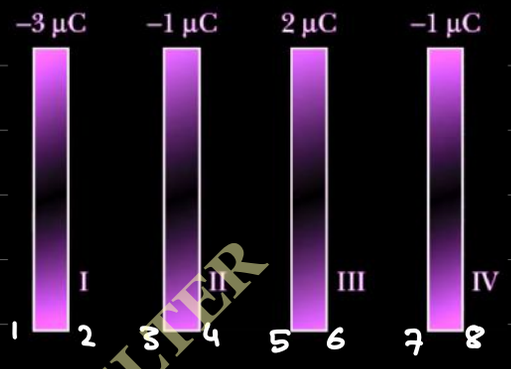
From ① & ②, $Q_1 = Q_8 = \frac{Q_A + Q_B + Q_C + Q_D}{2}$ QED.

Example: (Resnick Halliday Singhal):

Four large charged conducting sheets kept parallel to each other. Find the charge on each of the eight faces and electric field in region I, II, III, IV as shown in the figures below. (Area of each plate is $A = 1\ m^2$)



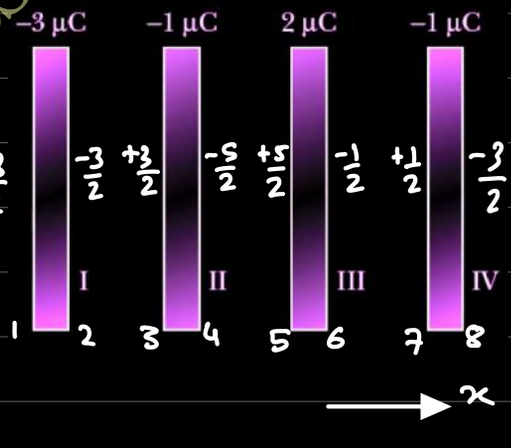
Solution: According to the theory in the previous section, $Q_1 = Q_8 = \frac{(-3 + -1 + 2 + -1)}{2}\ \mu C$

$\Rightarrow Q_1 = Q_8 = -3/2\ \mu C$.

Now charges on each plate can be found by charge conservation, and using the fact that facing faces will get equal and opposite charges. (Refer to the adjacent figure).

Now field in each region can be found using the fact that near a conductor field is given by $E = \sigma/\epsilon_0$ where σ is the local charge density. Thus:

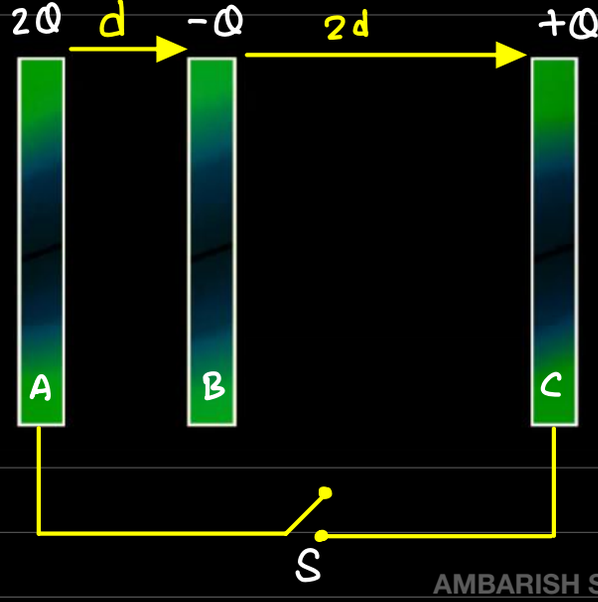
$$\vec{E_I} = \frac{Q_2}{A\epsilon_0}\hat{i} = \frac{-3}{2\epsilon_0}\ \mu N/C\ \hat{i}$$



Similarly $\vec{E_{II}} = \frac{-5}{2\epsilon_0}\ \mu N/C\ \hat{i}$, $\vec{E_{III}} = \frac{-1}{2\epsilon_0}\ \mu N/C$, $\vec{E_{IV}} = \frac{-3}{2\epsilon_0}\ \mu N/C$ Ans.

Example: System Of Parallel Plates With Shorting Switches:

Three plates A, B and C are given charges as shown in the figure. Separation between A & B is d and that between B & C is 2d. How much charge flows through the switch S after it is closed?



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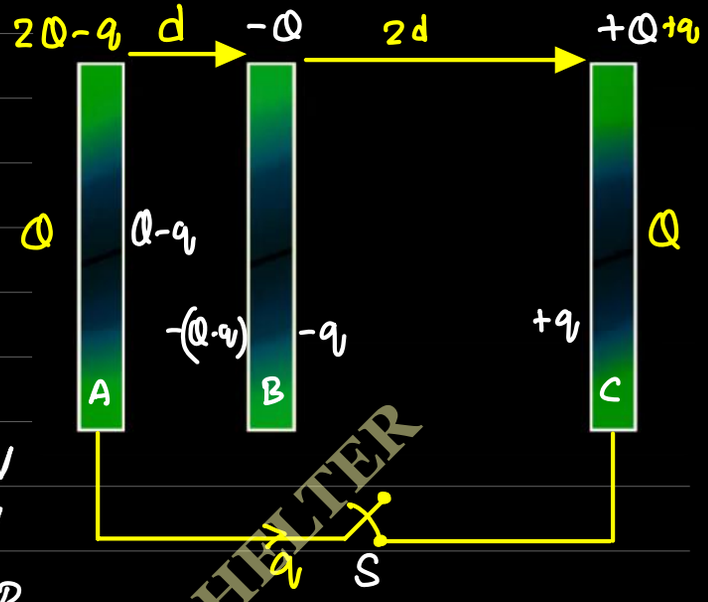
Solution: Whenever two plates are connected through a switch (shorted), their potentials become equal which essentially means that work done by electric field in going from one plate to the other becomes zero.



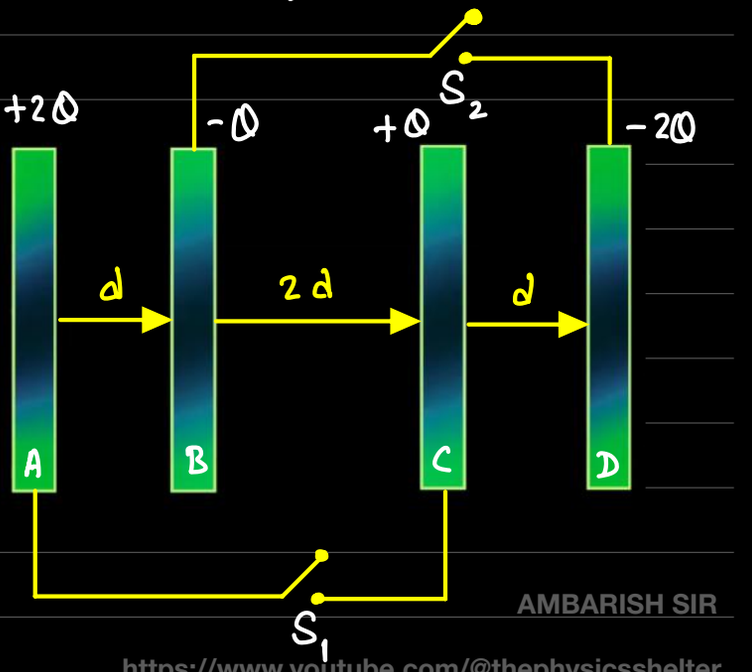
Now let a charge q flow through switch S. The extreme faces will get half the total charge of the system (which is still same as before), thus extreme faces get a charge Q each. Charges of other faces are represented in terms of variable q (Using charge conservation and the fact that facing faces get equal and opposite charges). Now field between A & B is $E_1 = \frac{Q-q}{A\epsilon_0}\hat{i}$ and that between B & C is $\frac{-q}{A\epsilon_0}\hat{i}$, since A & C are now equipotential $\therefore$ work done by electric field in going from A to C must be zero. Thus:

$$\left(\frac{Q-q}{A\epsilon_0}\hat{i}\right).d\hat{i} + \left(\frac{-q}{A\epsilon_0}\hat{i}\right).(2d\hat{i}) = 0$$

$$\Rightarrow \quad Q-q = 2q \Rightarrow q = Q/3.$$

Thus a total charge of $+Q/3$ flows from plate A to Plate C. (Actually this is equivalent to saying $-Q/3$ flows from C to A which is in a way more correct because actually electrons will flow).

EXAMPLE: Consider the system of four parallel plates as shown in the figure. The initial charges are indicated. Find the net charges flown through switches S_1 and S_2 after both of them are closed.



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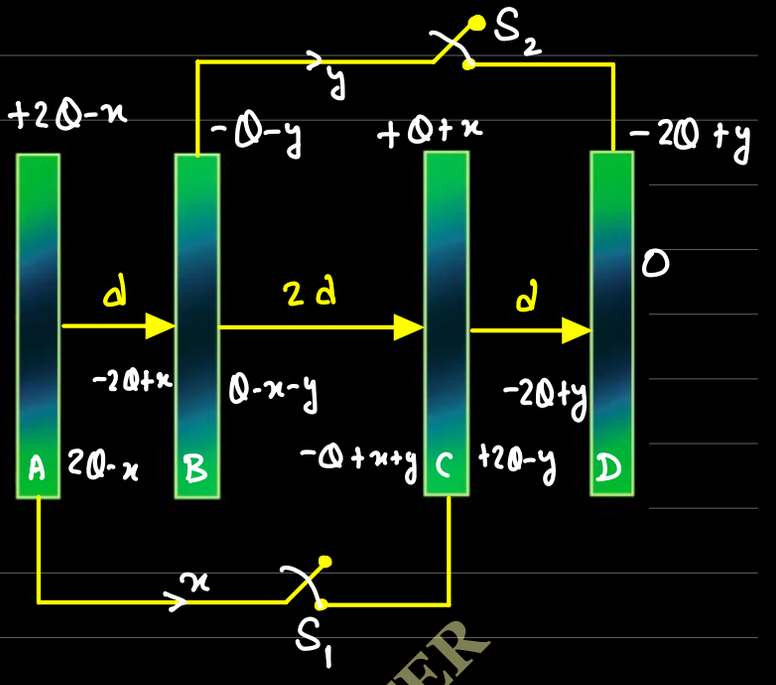
Solution: Let x and y be the charges flown through the switches. Charges on extreme must each be 0. Now by charge conservation and using the fact that facing faces get equal and opposite charges, we have shown the charges on each face in terms of variables x & y. Now we have two unknowns x & y and we can make two equations using the facts that plates A & C are at same potential and Plates B & D are at same potential (i.e work done by electric field in going from one plate to the other is zero).



Thus:

$$A \to C: \quad \frac{1}{A\epsilon_0}\left[(2Q-x)d + (Q-x-y)\times 2d\right] = 0 \qquad (1)$$

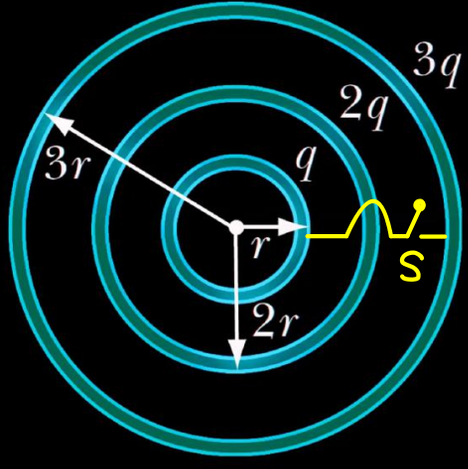
$$B \to D: \quad \frac{1}{A\epsilon_0}\left[(Q-x-y)\times 2d + (2Q-y)\times d\right] = 0 \qquad (2)$$

Solving (1) & (2) we get: $x = 4Q/5$, $y = 4Q/5$, thus charge flown through both switches is $Q/5$ each.

16. "Facing Faces" of Cylindrical / Spherical Shells Get Equal & Opposite Charges:

This is just a consequence of the property that surface of a cavity gets a charge equal in magnitude and opposite sign to the charge enclosed by the cavity. (Each shell is like a conductor with cavity).

Example Consider a system of three concentric shells, Find the charge flown through switch S after it is closed to short the innermost and outermost shells.



Solution: After closing the switch, the potential of inner shell will become equal to that of outer shell and we shall use this

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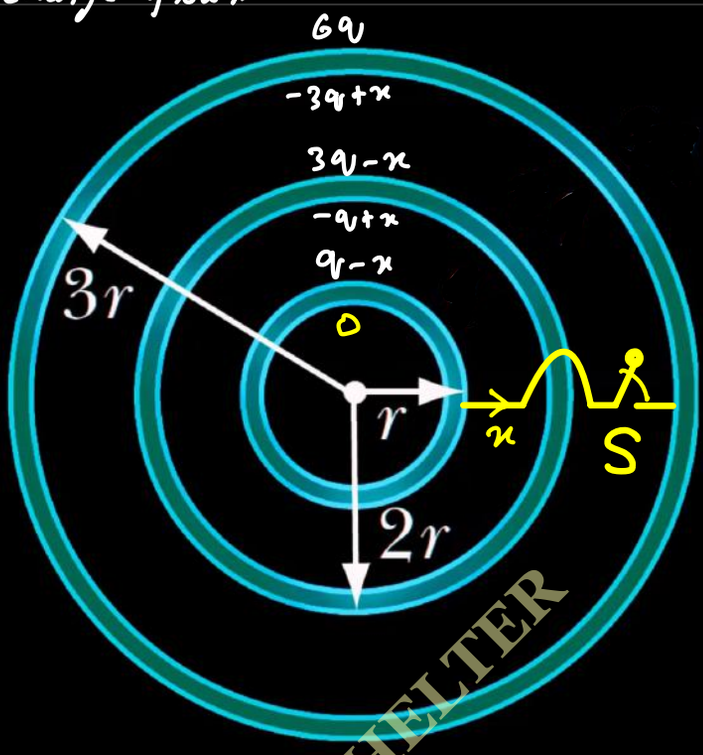
fact to solve for the unknown charge flown through the switch. Let charge flown through the switch be x then net charge on innermost shell becomes $q-x$ and that on outer shell becomes $3q+x$.



(All shells are thin so for calculating any potential we don't need to consider surface-wise charges, rather we can consider combined charge for both surfaces of inner and outer shells, had it been thick shells then we will need to consider charges surface-wise to calculate potentials).

Now let us equate potentials:

$$V_{in} = \frac{k(3q+x)}{3r} + \frac{k\times 2q}{2r} + \frac{k(q-x)}{r} = \frac{k\times 6q}{3r} = V_{out}. \quad (1)$$

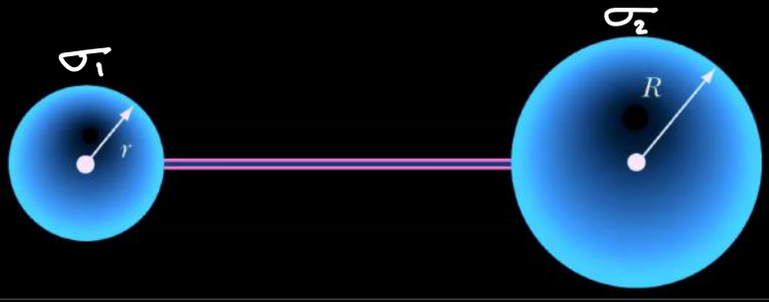
(Note for outer shell potential, radius for each charge is $3r$ ∵ outside or surface point of shell).

Now solving equation (1) for x: $x = 3q/2$. Ans.

Thus a total charge of $3q/2$ flows from inner shell to outer shell.

17. Small Radii Patches On An Isolated Conductor Have Higher Charge Densities:

To illustrate this fact we connect two conducting spheres with a very long conducting wire, so as to make the potentials of the spheres equal. (The whole system now behaves like a single conductor).



Now $V_1 = V_2 \Rightarrow \frac{1}{4\pi\varepsilon_0} \times \frac{\sigma_1 \times 4\pi r^2}{r} = \frac{1}{4\pi\varepsilon_0} \times \frac{\sigma_2 \times 4\pi R^2}{R}$

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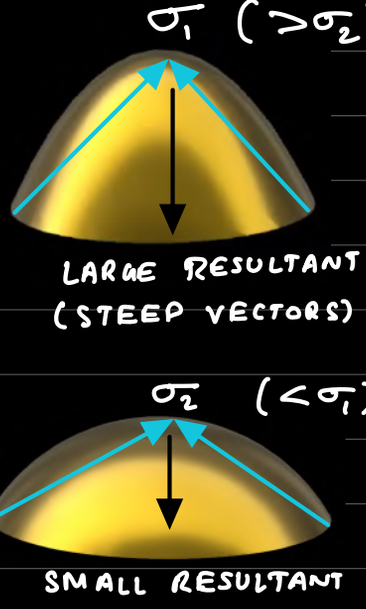
$\Rightarrow \sigma_1 r = \sigma_2 R$ or $\frac{\sigma_1}{\sigma_2} = \frac{R}{r}$ (1), Thus we see that the sphere with smaller radius of curvature has a larger density. Even though we have done our calculations for a special case, in general also we can say that convex patches of smaller radii of curvature will have larger charge densities.

→ Caution: It does not mean that concave patches will have charge densities of opposite polarity. Infact we have already shown that an isolated conductor cannot simultaneously have positive and negative charge densities.

→ The equation (1) is not quantitatively true, actual relation between charge density and Gaussian Curvature $\left(\frac{1}{R_1}+\frac{1}{R_2}\right)$ is complicated and far beyond scope of study at JEE level.

Understanding Charge Density Magnitude:



The adjacent figure shows two patches with different radii of curvature. It can be seen that where radius of curvature is small, the the field vectors from adjoining areas add up at a very small angle, thus giving a large resultant, which has to be cancelled by local patch (To make net field zero in metal metrix) thus due to small radius of curvature, σ_1 has to be large. At a place where radius of curvature is large, these vectors from surrounding yield a smaller resultant field and accordingly small charge density is required to cancel its field.

18. Corona Discharge As discussed in previous section, the regions of small radii of curvature get get greater charge density, thus if there are any pointed features on a conductor, they acquire very high charge density. Also recall that electric field in vicinity of

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a conducton is σ/ϵ_0, Thus at regions of small radii, electric field outside the conducton may become very high, which can ionise the surrounding air and hence make it conducting, in that case the charged conducton may start getting discharged by either gaining or losing electrons from surrounding air. This discharging is usually accompanied by energy dissipation in form of crackling sound and lightning, this phenomenon is called corona discharge.

LIGHTNING CONDUCTOR:

As discussed in previous section, the air near the pointed features on a conducton can become conducting due to high charge density. To prevent buildings from lightning we use this phenomenon. A pointed conducton is provided at the top of the buildings which is connected to earth on the other side. This provides a pathway to atmospheric lightning thus preventing the buildings from harm.

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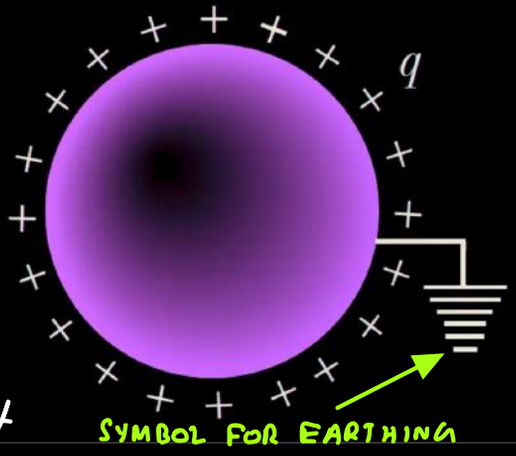
EARTHING: Recall that any conducton is an equipotential volume. The earth is also a conducton (due to various minerals and moisture present in it). Thus whenever a conducton is connected to earth through a connecting wire, its potential also becomes equal to earth's potential. Since earth is a very large sphere, therefore small amount of charge flowing in to earth or leaving from earth does not affect its potential. For purpose of calculations, conventionally assumed



to be a sphere of zero charge and hence zero potential. Thus whenever any conductor is "earthed", its potential also becomes zero.

→ Even though earth is assumed to be at zero potential wrt ∞, this is strictly not true, the earth has slight negative charge density on its surface, causing its potential to be substantially negative wrt ∞, however this should never be considered in calculations (This fact was not known at the time when concept of earthing was invented).

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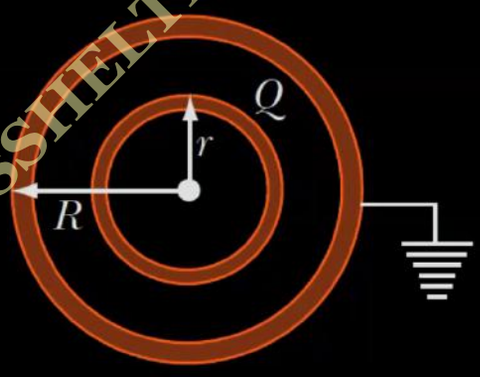
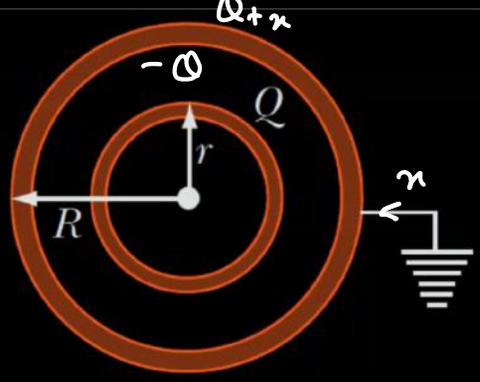
EXAMPLE: Assume that two spherical concentric conducting shells of radii r and R are kept as shown in the figure. The inner shell is given a charge Q and the outer shell is earthed. Find the charge supplied by the Earth and the potential of the inner shell.



Solution: Let a charge x flow from earth to the outer shell. ∵ the outer shell it earthed ∴ its potential must become zero Thus:

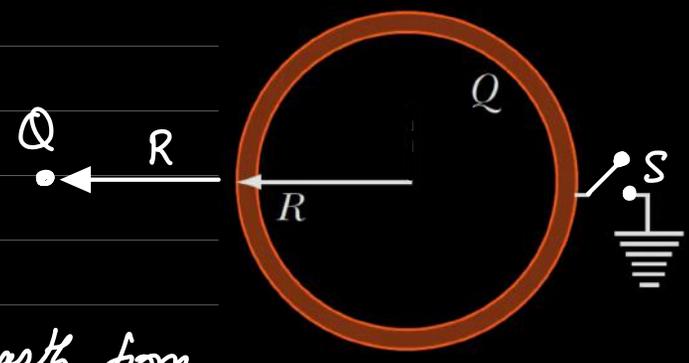
$$\frac{kQ}{R} + \frac{kx}{R} = 0 \Rightarrow x = -Q$$



Thus a charge $-Q$ flows in to the outer shell from earth.

→ We could also do this problem directly, ∵ potential of outer shell is zero ∴ there should be no work done in reaching it from ∞. ∴ field outside must be zero ∴ charge density on outer surface must be zero, and inner surface will get negative of the charge in its cavity.

Example : A shell having a net charge $+Q$ and a point charge $+Q$ are arranged as shown in the figure. Calculating the charge flown through the switch after it is closed.



Solution: Let a charge x flow in to earth from the shell, then final charge on shell becomes $Q-x$. Now since cavity has no field, $\therefore$ potential of shell is same as potential at the centre of its cavity, which should also be zero because of earthing. Thus equating potential at the centre to zero:

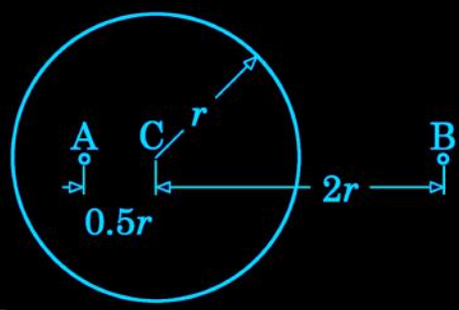
$$\frac{kQ}{2R} + \frac{k(Q-x)}{R} = 0 \quad \Rightarrow x = 3Q/2$$

Thus a total charge of $3Q/2$ flows from shell to the earth.

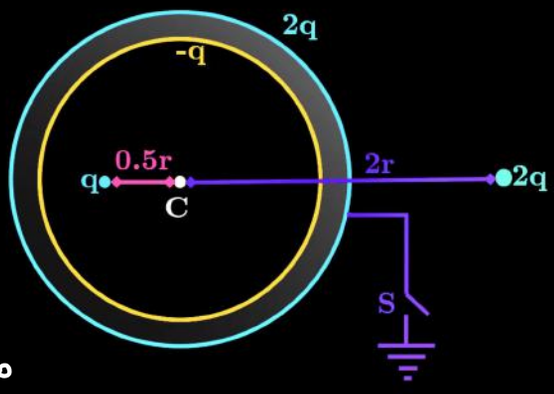
Example : Pathfinder MCQ 20 :

Consider a thin conducting shell of radius r carrying total charge q. Two point charges q and $2q$ are placed on points A and B, which are at distances $0.5r$ and $2r$ from the centre C of the shell respectively. If the shell is earthed, how much charge will flow to the earth?

(a) $2q$ (b) $3q$
(c) $4q$ (d) More than $2q$ and less then $3q$



Solution: Initial charge configuration is shown in the adjacent figure. Due to shielding property, the cavity volume and surface charges are as good as absent for contributing to potential of the shell. Thus for zero potential, the outer point charge $2q$ and the outer surface charge alone should also cause a zero potential. Also due to shielding, the potential due to outer surface charge of the shell and the external $2q$ at the centre must be same as their potential contribution to the surface C $\because$ these two together yield zero field everywhere in the cavity).



Now let a charge x flow in to earth upon closing the switch.

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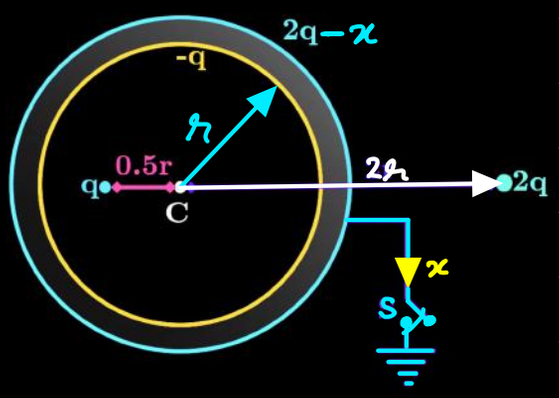
Thus the charge on outer surface will become $2q-x$, while other charges will remain as it is. Now calculating potential contribution of $2q-x$ and $2q$ at C:



$$\frac{k(2q-x)}{r} + \frac{k \times 2q}{2r} = 0$$

$$\Rightarrow x = 3q \text{ Ans.}$$

EXAMPLE : PATHFINDER : BUILD 33 :

Two metal spheres of radius r_1 and r_2 are connected by a thin conducting wire. The second sphere is surrounded by a grounded concentric conducting shell with uniform separation d between their facing surfaces. If the second sphere is given a charge Q, find charges acquired by all the three spheres. Assume that $d << r_2$, and the distance between the spheres is much larger than their radii.

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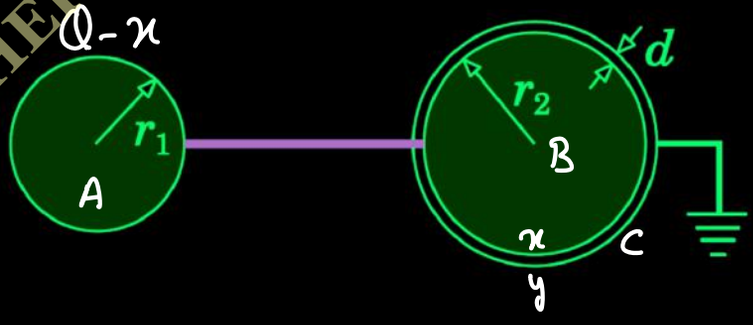
Solution: Let the charges on spheres be x, y and $Q-x$ as shown in the figure. We have here two unknowns x and y for which we will make the following equations:



① Equating potentials of sphere A and B

② Equating potential of C to zero.

Since the distance of A is very large from B & C ∴ the contribution to its potential due to B & C is negligible and vice-versa.

Now equating the potentials:

$$\frac{k(Q-x)}{r_1} = \frac{kx}{r_2} + \frac{ky}{r_2+d} \quad (1)$$

Now using earthing condition:

$$\frac{ky}{r_2+d} + \frac{kx}{r_2+d} = 0 \quad (2)$$

Solving for x & y, $x = Q\left(\frac{R_2(d+R_2)}{d(R_1+R_2)+R_2^2}\right)$, $y = \frac{-Q R_2(d+R_2)}{d(R_1+R_2)+R_2^2}$

and $Q-x = \frac{Q R_1 d}{d(R_1+R_2)+R_2^2}$

→ This is an accurate solution though does not match pathfinder solution. Now we will try approximations to match the pathfinder answer. ☺

Using binomial approximation in (1):

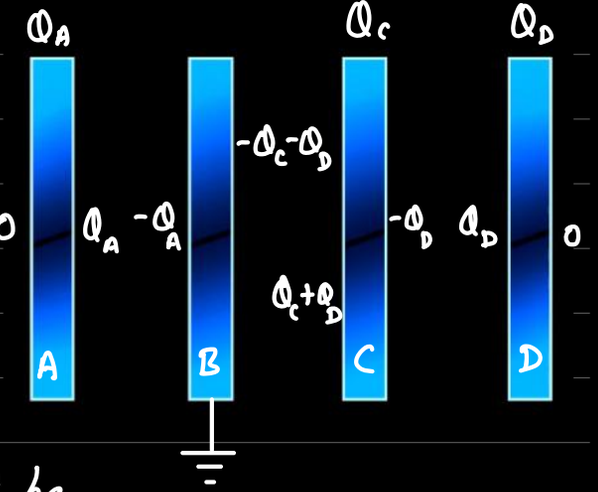
$$\frac{k(Q-x)}{R_1} = \frac{kx}{R_2} + \frac{ky}{R_2}\left(1-\frac{d}{R_2}\right) \quad (1')$$

also, (2) ⇒ $y = -x$ (2'), Now solving (1') and (2'):

$$x = \frac{Q R_2^2}{R_1 d + R_2^2}, \quad y = \frac{-Q R_2^2}{R_1 d + R_2^2} \text{ and } Q-x = \frac{Q d R_1}{R_1 d + R_2^2}$$

Earthing In A System Of Parallel Plates:

Earthing any one plate out of a system of charged parallel plates (treated as infinite) in general leads to a contradiction. For instance consider the system of parallel plates as shown in the figure. If the plate B is earthed, work done in bringing a unit charge from infinity to this plate must be zero, Now if outer surface of plate A (treated infinite) has any charge, then it will create a finite field all the way up to infinity and causing an infinite potential of plate A. Now work done in going from plate A to B is finite, thus plate B will also have to be at infinite potential. The other possibility is to assume a zero charge on outer surface of A. In that case, work done from infinity to plate A is zero, however still some work will be done from A to B, thus once again not making potential of B to be zero (Though still an improvement because



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taking finite charges on outer surface lead to infinite potential of earthed plate). The main reason for this contradiction is the modelling of finite plates (no matter how so large) as infinite plates. In practice, any finite plate can't give a constant field all the way to infinity and therefore must have a finite potential wrt infinity.

In practice, the outer surface of A will acquire some little amount of charge such that the work done from infinity to the outer surface of A is exactly negative of work done in going from inner surface of A to B. Now since comparatively very small charge on outer surface of A will do the job (satisfying zero potential at B), thus for practical purposes, we assume this charge to be zero. Now we can state our practical problem solving rule as follows:

→ In a system of parallel plates, if any of the plates is earthed, the extreme surfaces of the system will have zero charges (to maintain finite potentials) and hence net charge of the system will also be zero (∵ facing faces anyway add up to zero).

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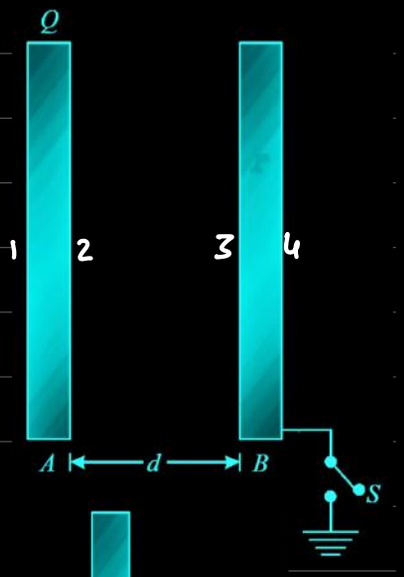
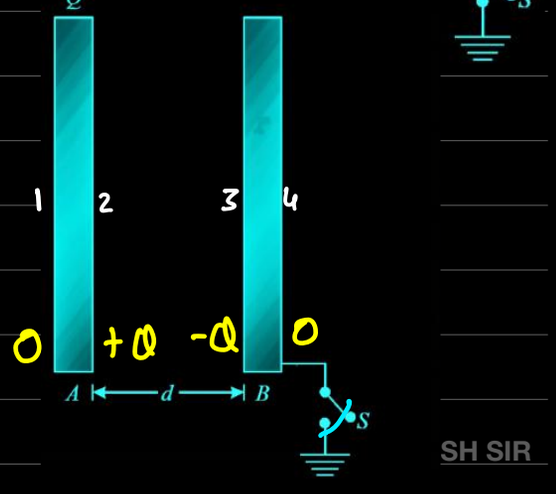
EXAMPLE : Find charges appearing on each of the faces 1, 2, 3 and 4 after the earthing switch is closed.



Solution: As discussed in the previous section, extreme faces must get a zero charge upon earthing, thus $Q_1 = Q_4 = 0$.
Now remaining charges can be found by charge conservation and using the fact that facing faces get equal and opposite charges. Thus, $Q_2 = Q$, $Q_3 = -Q$. The appearing charges are shown in the adjacent figure.



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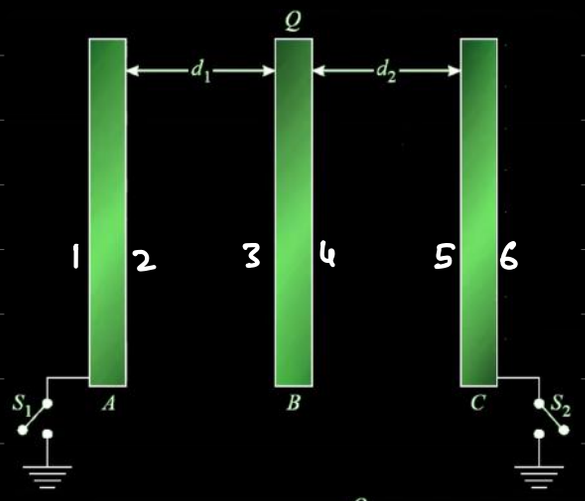
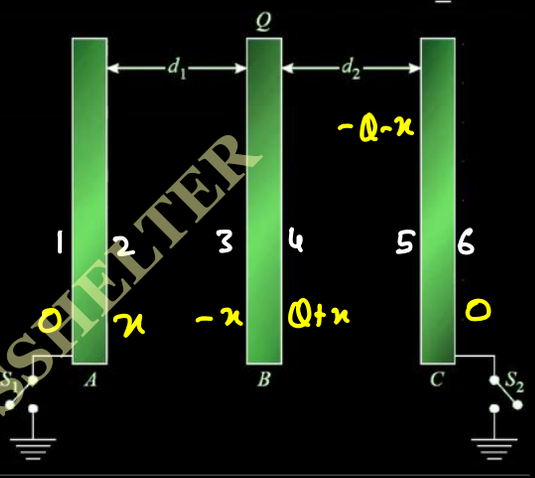
EXAMPLE: Find the final charges on each face 1, 2, 3, 4, 5 and 6 after both earthing switches are closed, also find the charges flown through switches.



Solution: Extreme faces must each get a zero charge as discussed earlier.
Now we shall use the fact that facing faces get equal and opposite charges and also the fact that earthed plates have same potential (∵ work done in going from one earthed plate to the other must be zero). Refer to the adjacent figure.



$$\frac{1}{A\epsilon_0}\left[x \times d_1 + (Q+x) \times d_2\right] = 0$$

$$\Rightarrow \quad x = -Q\left[\frac{d_2}{d_1+d_2}\right], \quad Q+x = +Q\left[\frac{d_1}{d_1+d_2}\right]$$

Now charge flown through $S_1 = x = -Q\left[\frac{d_2}{d_1+d_2}\right]$ Ans.

charge flown through $S_2 = Q+x = +Q\left[\frac{d_1}{d_1+d_2}\right]$ Ans

EXAMPLE: 200 MORE PUZZLING: 148

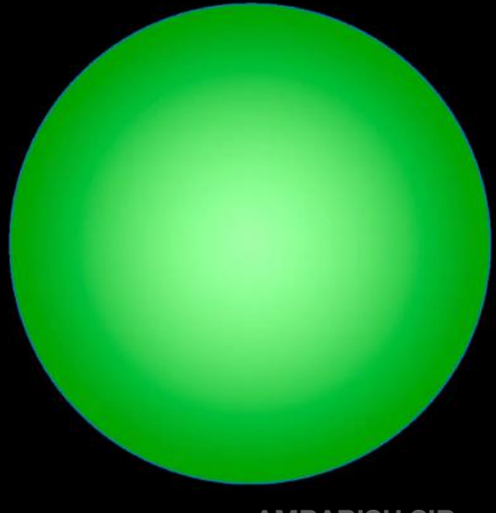
P148** A thin metal disc of radius R is charged, and carries a total electric charge of Q. Find the charge distribution on the disc.

Solution: This problem is quite tricky. Here the mathematical conditions that we need to satisfy are:

① $\int \sigma \, dS = Q$

② Field should be zero everywhere in the plane of the disc.

It turns out that both the conditions can be satisfied by relating the problem to the field of



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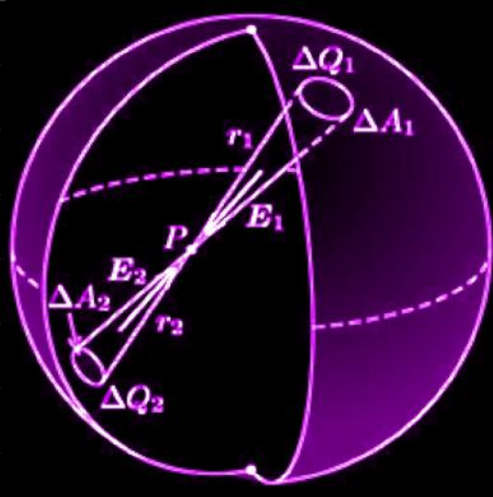
a uniformly charged spherical shell. Consider a point P on the imaginary equatorial disc of the uniformly charged spherical shell. Recall the alternatin proof of shell theorem. Here the field due to ΔA_1 is cancelled by the field due to ΔA_2 (i.e $\frac{Q_1}{r_1^2} = \frac{Q_2}{r_2^2}$). Now suppose



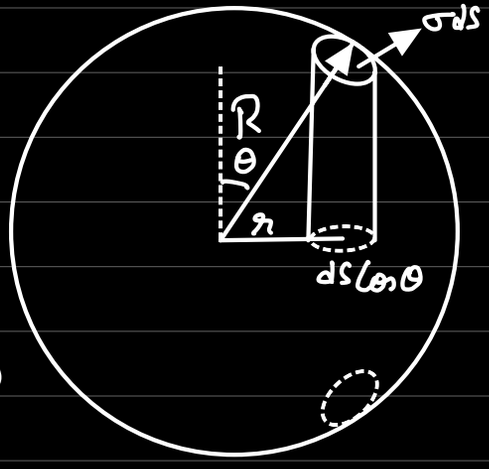
we collapse these areas on to the equatorial plane to respective area $\Delta A_1'$ and $\Delta A_2'$ respectively. Then the magnitude of their respectin fields at P would be, $\frac{Q_1}{d_1^2}$ and $\frac{Q_2}{d_2^2}$

(Say E_1' & E_2' respectively)

Now $\because \Delta MPM' \sim \Delta NPN' \therefore \frac{d_1^2}{d_2^2} = \frac{r_1^2}{r_2^2} = C$ (say)

$\therefore E_1' = \frac{Q_1}{C r_1^2}$ and $E_2' = \frac{Q_2}{C r_2^2}$,

Now Since $\frac{Q_1}{r_1^2} = \frac{Q_2}{r_2^2} \therefore E_1' = E_2' \therefore$ once again these charges would create a cancelling field in the plane of the disc. By a similar logic we can say, net field in the plane of the disc would be zero by collapsing all the pairwise cancelling patches of shell. This collapsing also automatically satisfies the charge conservation. Now Consider a surface patch at polar angle θ and surface area dS. This patch collapses to an area $dS\cos\theta$. Similarly its mirror image patch in "Southern" hemisphere will collapse on to same $dS\cos\theta$. Thus required charge density is:



$$\sigma' = \frac{2\sigma dS}{dS\cos\theta} = 2\sigma\sec\theta$$

Now $\sin\theta = \frac{r}{R} \Rightarrow \sec\theta = \frac{R}{\sqrt{R^2 - r^2}}$ and $\sigma = \frac{Q}{4\pi R^2}$

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$\Rightarrow$ required charge density, $\sigma' = 2 \times \frac{Q}{4\pi R^2} \times \frac{R}{\sqrt{R^2 - x^2}}$

$\Rightarrow \quad \sigma' = \frac{Q}{2\pi R\sqrt{R^2 - x^2}}$ Ans.

EXAMPLE :

Consider a random shaped earthed conductor as shown in the figure.

(a) Can this conductor have simultaneous positive and negative local charge densities

(b) Can the total induced charge on surface have a magnitude greater than Q.

Solution: (a) If the conductor gets a positive charge density anywhere then, let us consider if field lines from it can go anywhere:

(I) They can't go to infinity (otherwise the conductor won't be at zero potential unt infinity)

(II) They can't obviously terminate at $+Q$. Thus such field lines can't go anywhere $\therefore$ simultaneous negative and positive charge density is ruled out. There will be only negative charge density.

(b) Recall that number of field lines emanating / terminating on a charged body is proportional to its charge. If the body has a charge $-q$ such that $q > Q$, then obviously not all field lines terminating in $-q$ can be sourced from $+Q$. $\therefore$ Some of them will have to be sourced from ∞. Once again it will lead to a contradiction. Because if some field lines from infinity terminate on conductor then, the conductor can't be at zero potential.

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Example : 200 More Puzzling : 150:

P150* A solid metal sphere of radius R is divided into two parts by a planar cut, made in such a way that the outer surface of the smaller part of the sphere is πR^2. The cut surfaces are coated with a negligibly thin insulating layer, and the two parts are put together again, so that the original shape of the sphere is restored. Initially the sphere is electrically neutral.

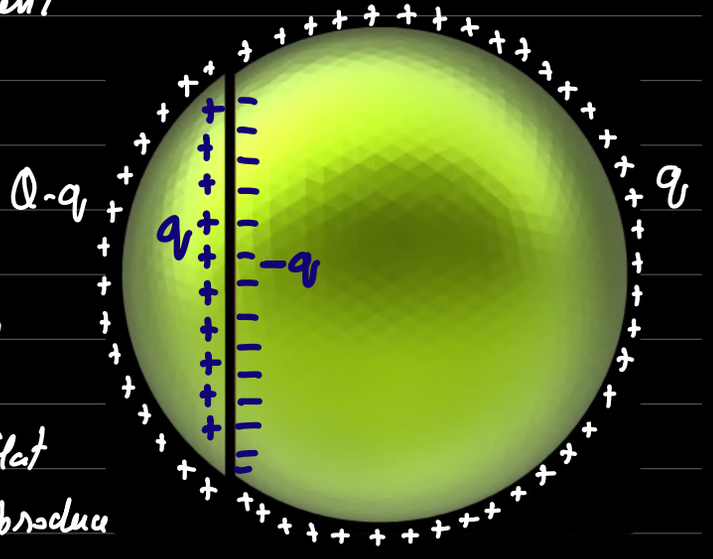
The smaller part of the sphere is now given a small positive electric charge $+Q$, while the larger part of the sphere remains neutral. Find

a) the charge distribution throughout the sphere,

b) the electrostatic interaction force between the two pieces of the sphere.

Solution: A solid metallic body is as good as a metallic shell with an unoccupied cavity for the purpose of field and charge distribution equivalence. Thus for calculation purposes we may consider the two parts of the sphere as shells in shape of segments of a sphere. Now we know that net field in both cavities must be zero. So we need to guess a charge configuration which makes net field inside both cavities zero subject to the constraint of charge conservation. If we are able to guess one such configuration, we can be sure that that is the only such configuration because of uniqueness theorem. Now let us try to guess such a configuration:

Consider the configuration shown in adjacent figure, if the outer surfaces of both shells get same charge density and the flat surfaces get equal and opposite uniform charges then using shell theorem it can be readily seen that zero field condition is satisfied in both cavities. (Note that charges on flat surfaces are so close together that they produce a net field zero everywhere in space. Now our task remains to equalise charge densities of curved portions of spherical sector surfaces and we are done. Thus:



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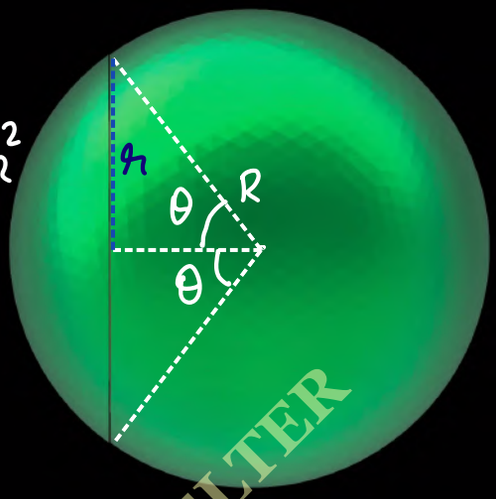
$$\sigma = \frac{Q-q}{\pi R^2} = \frac{q}{3\pi R^2} \Rightarrow q = \frac{3}{4}Q \quad \text{Ans.}$$

(Now we know q and entire charge distribution in terms of q).

Now Solid angle of the smaller cap:

$$\omega = \pi R^2/R^2 = \pi = 2\pi(1-\cos\theta)$$

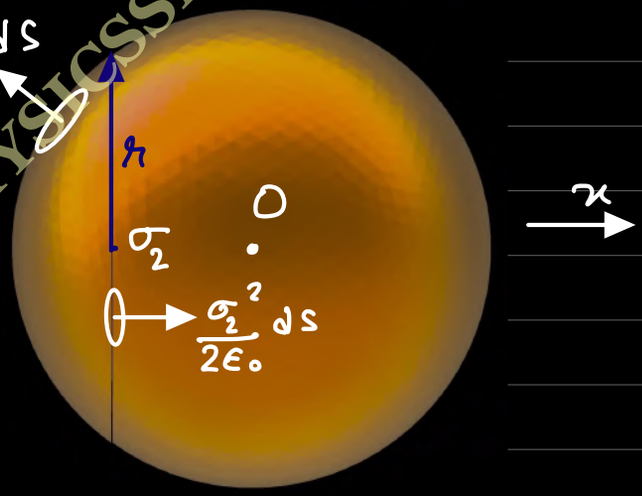
$$\Rightarrow \theta = 60^\circ \Rightarrow r = R\sin 60^\circ = R\frac{\sqrt{3}}{2}.$$



Now net force on smaller section can be found using concept of electrostatic pressure and projection area.

Thus required force:

$$F = \left(\frac{\sigma_2^2}{2\epsilon_0} \times \pi r^2 - \frac{\sigma_1^2}{2\epsilon_0} \times \pi r^2\right)\hat{i}$$



where $\sigma_2 = \frac{q}{\pi r^2}$, $\sigma_1 = \frac{Q-q}{\pi R^2}$

Substituting the values, we get:

$$\vec{F} = \frac{45Q^2}{128\pi R^2\epsilon_0}\hat{i} \quad \text{Ans.}$$

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